

# Exact algebraic confluence for torus one-point Virasoro-block recursions

LESLIE P. POLZER

Independent Researcher

polzer@fastmail.com

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**Abstract.** Fix a nonzero algebraic Coulomb-gas parameter  $b_*$ . We prove an exact, finite confluence formula for every coefficient of the eta-reduced torus one-point Virasoro block at  $c_* = 1 + 6(b_* + b_*^{-1})^2$ . Generic Kac-recursion terms are grouped by their common limiting internal weight. Each coefficient of the resulting fixed-central-charge principal part is a collision moment: the constant term of a finite sum of Laurent germs in  $t = b - b_*$ . Separated pole sets make every collision class individually regular, while a valuation majorant and a backward precision lemma give an a priori finite jet budget. Uniqueness of partial fractions identifies the construction coefficientwise in  $q$  with the inverse-Shapovalov definition of the meromorphic Verma block. The formal germ is an exact algebraic device, not a numerical regularization or an intrinsic recursion in the specialized representation category. At the Ising charge we classify the collision classes explicitly and use local Smith and crossing forms to interpret the first two embedding diamonds, including the first double pole at level seven and the level-ten spin example. Ancillary scripts replay the finite computations in exact rational arithmetic.

**Keywords.** Virasoro algebra; conformal blocks; torus one-point block; Zamolodchikov recursion; minimal models; Ising model.

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## 1. Introduction

Residue recursions for Virasoro blocks originate with Zamolodchikov’s recursive representations of the sphere four-point block [Zam84, Zam87]; see [CCY19] for a modern general treatment. For generic central charge, the eta-reduced torus one-point Virasoro block obeys the residue recursion conjectured by Poghossian [Pog09] and derived in [HJS10, Section 3, equations (7)–(9)]. At rational values of the Coulomb-gas parameter, distinct Kac labels have the same degenerate weight and the individual generic-recursion summands develop parameter poles. Javerzat, Santachiara, and Foda analyzed the cancellation of these non-physical poles in examples and in the identity-insertion character problem [JSF18, Sections 4–6, especially equations (60)–(61)]. Stocco subsequently constructed central-charge-pole-free torus expressions through fourth order and proposed an all-order Laurent-expansion procedure,

whose cancellation and algebraic form were left conjectural [Sto22, Sections II.2–II.3 and Appendix A]. Ribault identifies a recursion at rational  $\beta^2$  as an open problem and singles out torus one-point Virasoro blocks as its simplest incarnation [Rib26, Section 4.3.3]. Higher-order internal-weight poles and rational limits of sphere four-point blocks were also analyzed by Ribault [Rib19, Section 3]; those results provide important precedent for the confluence mechanism, although they do not give a finite torus one-point recursion.

This paper solves a precise coefficientwise version of that problem at every nonzero algebraic value  $b_*$  of the Coulomb-gas parameter. The object is the meromorphic Verma-module block, and the conclusion is coefficientwise in  $q$ . At every fixed level we partition the generic Kac summands by their common limiting weight at  $b = b_*$ . Each collision class has a regular sum, and every coefficient of its internal-weight principal part is a finite constant term of finite Laurent sums. The shifted recursion decreases the remaining level, while a recursive valuation majorant supplies an a priori finite jet budget.

The construction removes the divergent, non-physical poles in the deformation parameter; the genuine Laurent poles in the internal weight  $\Delta$  remain. It uses the formal germ  $b = b_* + t$  to combine generic summands before taking a constant term. Thus it gives an exact formula at fixed  $c_*$ , with no numerical  $\varepsilon$ -limit, but it is not an intrinsic recursion formulated solely in the specialized representation category. Nor does it by itself construct irreducible minimal-model traces at degenerate internal weights.

In the notation of Sections 4–7, the main result reads: for every level  $N \geq 1$ ,

$$H_N(c_*, \Delta, d) = \sum_{\lambda \in \Lambda_N(b_*)} \sum_{k=1}^{M_{\lambda,N}^*} \frac{A_{\lambda,k,N}^*(d)}{(\Delta - \lambda)^k}, \quad (1.1)$$

$$A_{\lambda,k,N}^*(d) = \text{CT}_t \sum_{a \in \mathcal{C}_{\lambda,N}^*} R_a(t) S_{a,N-\ell_a}(t) (h_a(t) - \lambda)^{k-1}.$$

where  $H_N$  is the  $N$ -th eta-reduced block coefficient,  $\Lambda_N(b_*)$  is the finite set of limiting Kac weights,  $\mathcal{C}_{\lambda,N}^*$  is the finite class of labels  $a = (r, s)$  with  $rs \leq N$  that collide at  $\lambda$ ,  $t = b - b_*$  is the exact Coulomb-gas germ, and  $R_a$  is the generic residue coefficient of the label  $a$ ,  $S_{a,M}$  is the shifted Laurent block of Definition 7.1,  $\text{CT}_t$  extracts the constant term of a Laurent germ, and  $M_{\lambda,N}^*$  is the finite structural pole bound of Definition 7.3. Every ingredient is a finite jet of an explicit rational function of  $b$ . Theorem 4.4 proves this identity: the constant-term moments on the right are exactly the partial-fraction coefficients of the inverse-Shapovalov block, at every power of  $q$ . At minimal-model values it applies directly, and the collision classes are the elementary Diophantine classes of Proposition 3.2. Theorem 7.6 records the Ising specialization, where the coefficient field improves from  $\mathbb{Q}(b_*)$  to  $\mathbb{Q}$ .

The local calculations exhibit how the algebraic confluence records Virasoro embedding geometry. The earliest collision is the transverse energy-module collision at level three; its two branches first meet at level seven and create the first double pole. The fixed- $c$  formulas are Theorems 5.8 and 8.3. The spin-module collision at level four and its larger level-ten diamond are treated in Theorems 5.11 and 9.4. Finally, Theorem 11.3 distinguishes the Verma identity specialization from the irreducible quotient (Proposition 11.1) and records

the three standard Ising characters as a separate BGG consistency check.

The paper is organized as follows. Section 2 defines the block and its eta reduction and collects notation; the symbol families are indexed in Remarks 2.5 and 2.6. Section 3 classifies the minimal-model collision classes and records the Ising specialization. Section 4 proves the finite coefficientwise partial-fraction expansion and states the main theorem before the examples. Section 5 develops the local Smith and crossing-form toolkit and computes the transverse collisions at levels three and four. Section 6 recalls the generic residue data and proves the generic torus recursion. Section 7 proves the general finite confluence theorem (Theorem 4.4) and its Ising corollary. Sections 8 and 9 work out the two embedding diamonds—the energy diamond with its first double pole at level seven, and the spin diamond at level ten; Corollary 9.5 identifies these local computations with the corresponding collision moments, and Table 1 summarizes them. Section 10 summarizes the output through level four and compares it with the expressions of Stocco; the expanded polynomials are collected in Appendix A. Section 11 treats the irreducible specialization and the Ising characters; and Section 12 discusses scope and open directions.

A word on the status of computations. Several local statements rest on finite exact-rational calculations: PBW commutation, Ward recursion, determinants, kernels, and crossing forms. Every such calculation is replayed in exact arithmetic by the ancillary audit scripts described in Remark 9.6 and in the Code availability section; no step of any proof depends on floating-point arithmetic. The one quantity not reconstructed symbolically by an independent script is the level-ten second crossing form itself; its scalar  $\gamma_{10}$  is obtained in Theorem 9.3 by comparison with the confluence and is verified directly against the inverse-Shapovalov definition by `tools/audit_grade10_direct.py`.

## 2. Blocks and the reduced series

**Definition 2.1** (Virasoro Verma module). The Virasoro algebra has generators  $L_n$ ,  $n \in \mathbb{Z}$ , and central element  $c$ , with

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

The Verma module  $M(c, \Delta)$  is generated by a vector  $|\Delta\rangle$  such that

$$L_0|\Delta\rangle = \Delta|\Delta\rangle, \quad L_n|\Delta\rangle = 0 \quad (n > 0).$$

For a partition  $Y = (y_1 \geq \dots \geq y_\ell)$ , set  $L_{-Y} = L_{-y_1} \cdots L_{-y_\ell}$ ,  $|Y| = \sum_i y_i$ , and  $\ell(Y) = \ell$ .

**Definition 2.2** (Shapovalov and torus matrices). At level  $N$ , let

$$G_N(c, \Delta)_{Y,Y'} = \langle \Delta | L_Y L_{-Y'} | \Delta \rangle, \quad |Y| = |Y'| = N,$$

where  $L_Y = L_{y_\ell} \cdots L_{y_1}$ . For a primary field  $V_d(z)$  normalized by  $\langle \Delta | V_d(z) | \Delta \rangle = z^{-d}$ , let

$$\rho_N(c, \Delta, d)_{Y,Y'} = \langle \Delta | L_Y V_d(1) L_{-Y'} | \Delta \rangle.$$

Whenever  $G_N$  is invertible, define

$$F_N(c, \Delta, d) = \sum_{|Y|=|Y'|=N} (G_N(c, \Delta)^{-1})_{Y, Y'} \rho_N(c, \Delta, d)_{Y, Y'}.$$

**Definition 2.3** (Eta-reduced block). Write

$$\mathcal{F}_\Delta^{(c)}(d \mid q) = q^{\Delta - c/24} \sum_{N \geq 0} F_N(c, \Delta, d) q^N$$

and define

$$H_\Delta^{(c)}(d \mid q) = \prod_{m \geq 1} (1 - q^m) \sum_{N \geq 0} F_N(c, \Delta, d) q^N = \sum_{N \geq 0} H_N(c, \Delta, d) q^N.$$

**Lemma 2.4** (*Polynomial matrix entries*). For fixed  $N$ , every entry of  $G_N(c, \Delta)$  belongs to  $\mathbb{Q}[c, \Delta]$ , and every entry of  $\rho_N(c, \Delta, d)$  belongs to  $\mathbb{Q}[c, \Delta, d]$ .

*Proof.* Repeated use of the Virasoro commutator moves every positive mode in  $L_Y$  to the right. Positive modes annihilate  $|\Delta\rangle$ , while each  $L_0$  and the central element act by  $\Delta$  and  $c$ , respectively. This gives the first assertion.

For the second assertion use, in addition,

$$[L_n, V_d(z)] = z^n (z \partial_z + (n+1)d) V_d(z).$$

Move positive modes through  $V_d(z)$ , then move the remaining negative modes through  $V_d(z)$  toward the bra. The process terminates because each commutation removes one mode from one side of the insertion. The remaining derivatives act on  $z^{-d}$ ; after setting  $z = 1$ , their values are polynomials in  $d$ . The other commutators contribute only polynomials in  $c$  and  $\Delta$ . ■

*Remark 2.5* (Terminology and notation). The  $L_0$ -offset  $N$  of a graded component is called its *level*. For a positive Kac label  $a = (r, s)$ , we write  $\ell_a = rs$  for the level of its singular vector. By Lemma 2.4 and Cramer's rule, each  $F_N$  and  $H_N$  is a rational function of  $(c, \Delta)$  and a polynomial in  $d$ ; note  $H_0 = 1$ . For a rational function  $f$  of  $\Delta$  and a point  $\lambda$ , we write  $\text{PP}_{\Delta=\lambda} f$  for the principal part of  $f$  at  $\lambda$ , that is, the sum of the negative-power terms of its Laurent expansion, and  $\text{Res}_{\Delta=\lambda} f$  for the coefficient of  $(\Delta - \lambda)^{-1}$ .

*Remark 2.6* (Symbol families). For reference, the following letters serve more than one family, always distinguished by their subscripts and context. The letter  $e_i$  denotes local Smith exponents (Section 5);  $e_j$ , used only inside the proofs of Lemma 4.1 and Proposition 6.4, denotes the coefficients of the Euler product  $\prod_{m \geq 1} (1 - q^m)$ ;  $e = (u, v)$  is a Kac label in Section 7; and  $\varepsilon_a = \text{ord}_t \delta_a$  is the collision order of Definition 7.3. The calligraphic  $\mathcal{A}_{r,s}, \mathcal{P}_{r,s}, \mathcal{R}_{r,s}$  are the generic residue data of Definition 6.1, whereas  $A_{\lambda,k,N}^*$  are the fixed- $c_*$  partial-fraction coefficients of Theorem 4.2 and  $\tilde{A}_{\lambda,k,N}^*$  the collision moments of Definition 7.3; the two are identified in Theorem 4.4. Their Ising aliases  $A_{n,k,N}$  and  $\tilde{A}_{n,k,N}$  are used after Theorem 7.6.  $\rho_N$  is the torus matrix of Definition 2.2, while  $\rho_e$  in Lemma 7.2 is a valuation bound; similarly  $p(N)$  counts partitions, while  $p_{a,e}$  in Lemma 7.4 is a precision

offset. Finally,  $b$  is the Coulomb-gas parameter,  $b_\chi$  the crossing scalar of Lemma 5.6, and  $\beta$  a crossing form; Stocco's  $\beta = b^2$  occurs only in Remark 10.1. In the proofs of Theorems 8.3 and 9.4, the letters  $\alpha, \beta, \mu, \nu$  abbreviate the four Kac labels of the corresponding embedding diamond and denote different labels in the two proofs; in particular,  $\beta$  is there a Kac label, not a crossing form. Finally, the torus elements  $E_2, E_3$  of Proposition 5.7 are indexed by the levels of the parent singular vectors  $\xi_2, \xi_3$ , whereas  $E_4, E_5$  of Theorem 8.3 are indexed by the relative levels of the child vectors  $\phi_4, \phi_5$ .

### 3. Algebraic specializations and Ising resonance classes

**Definition 3.1** (Kac weights and fixed algebraic specializations). For  $b \in \mathbb{C}^\times$ , set

$$c(b) = 1 + 6(b + b^{-1})^2, \quad h_{r,s}(b) = \frac{(b + b^{-1})^2 - (rb + sb^{-1})^2}{4}.$$

Thus  $h_{r,s}(b) = \Delta_{r,s}(c(b))$  is the standard Kac degenerate weight. Fix an algebraic number  $b_\star \in \overline{\mathbb{Q}}^\times$ , and put

$$c_\star = c(b_\star), \quad \Lambda_N(b_\star) = \{h_{r,s}(b_\star) : r, s > 0, rs \leq N\}.$$

For  $\lambda \in \Lambda_N(b_\star)$ , define the finite collision class

$$\mathcal{C}_{\lambda,N}^\star = \{(r, s) \in \mathbb{Z}_{>0}^2 : rs \leq N, h_{r,s}(b_\star) = \lambda\}.$$

For the Ising specialization, write

$$b_0 = \frac{2i}{\sqrt{3}}, \quad c(b_0) = \frac{1}{2}.$$

For  $r, s \in \mathbb{Z}_{>0}$ , define

$$\Delta_{r,s} = \frac{(4r - 3s)^2 - 1}{48}.$$

Thus  $\Delta_{r,s} = h_{r,s}(b_0)$ . Set

$$\lambda_n = \frac{n^2 - 1}{48} \quad (n \in \mathbb{Z}_{\geq 0}).$$

**Proposition 3.2** (*Minimal-model collision classes*). Let  $p, p'$  be coprime positive integers and choose  $b_\star^2 = -p'/p$ . Then

$$c_\star = 1 - \frac{6(p' - p)^2}{pp'}, \quad h_{r,s}(b_\star) = \frac{(p'r - ps)^2 - (p' - p)^2}{4pp'}.$$

For positive pairs  $(r, s)$  and  $(r', s')$ , their specialized weights are equal if and only if, for some  $\ell \in \mathbb{Z}$ ,

$$(r', s') = (r + p\ell, s + p'\ell) \quad \text{or} \quad (r', s') = (p\ell - r, p'\ell - s).$$

*Proof.* The displayed formulas follow by substituting  $b_\star^2 = -p'/p$  in Definition 3.1. Equality of the two weights is equivalent to

$$p'r' - ps' = \pm(p'r - ps).$$

For the plus sign, coprimality gives  $(r' - r, s' - s) = (p\ell, p'\ell)$ ; for the minus sign it gives  $(r' + r, s' + s) = (p\ell, p'\ell)$ . The converse is immediate. ■

**Proposition 3.3** (*Exact collision classification*). *For positive pairs  $(r, s)$  and  $(r', s')$ ,  $\Delta_{r,s} = \Delta_{r',s'}$  if and only if one of the following holds for some  $\ell \in \mathbb{Z}$ :*

$$(r', s') = (r + 3\ell, s + 4\ell),$$

or

$$(r', s') = (3\ell - r, 4\ell - s).$$

*Proof.* The equality of the two weights is equivalent to

$$4r' - 3s' = \pm(4r - 3s).$$

If the sign is  $+$ , then  $4(r' - r) = 3(s' - s)$ . Since  $\gcd(3, 4) = 1$ , there is an  $\ell \in \mathbb{Z}$  with  $(r' - r, s' - s) = (3\ell, 4\ell)$ . If the sign is  $-$ , then  $4(r' + r) = 3(s' + s)$ , and the same coprimality argument gives  $(r' + r, s' + s) = (3\ell, 4\ell)$ . The converse follows by direct substitution. ■

**Corollary 3.4** (*Distinct weights and local finiteness*). *The set of distinct positive-label degenerate weights at  $c = 1/2$  is*

$$\{\lambda_n : n \in \mathbb{Z}_{\geq 0}\}.$$

*Every collision class contains infinitely many positive pairs. For each fixed level  $N$ , only finitely many classes contain a pair with  $rs \leq N$ .*

*Proof.* For every positive pair,  $\Delta_{r,s} = \lambda_{|4r-3s|}$ . Conversely, given  $n \geq 0$ , choose a sufficiently large positive  $r \equiv n \pmod{3}$ ; then  $s = (4r - n)/3$  is a positive integer and  $4r - 3s = n$ . Starting with any positive pair in a class, the pairs  $(r + 3\ell, s + 4\ell)$ ,  $\ell \geq 0$ , are positive and remain in the same class. Finally,  $rs \leq N$  implies  $r \leq N$  and  $s \leq N$ , so only finitely many pairs, and hence classes, occur. ■

## 4. Coefficientwise partial fractions

**Lemma 4.1** (*Large internal weight*). *For fixed  $c, d, N$ ,*

$$\lim_{\Delta \rightarrow \infty} F_N(c, \Delta, d) = p(N),$$

where  $p(N)$  is the number of partitions of  $N$ . Consequently,

$$\lim_{\Delta \rightarrow \infty} H_N(c, \Delta, d) = \delta_{N,0}.$$

*Proof.* For a partition  $Y$ , put  $a_Y = \ell(Y)$ , and rescale the PBW vector  $L_{-Y}|\Delta\rangle$  by  $\Delta^{-a_Y/2}$ . Commuting positive modes to the right shows that

$$\deg_{\Delta} G_N(Y, Y') \leq \min(a_Y, a_{Y'}).$$

If  $a_Y = a_{Y'}$  and  $Y \neq Y'$ , the degree is at most  $a_Y - 1$ . For  $Y = Y'$ , the leading term is

$$\left( \prod_{j \geq 1} m_j(Y)! (2j)^{m_j(Y)} \right) \Delta^{a_Y},$$

where  $m_j(Y)$  is the multiplicity of  $j$  in  $Y$ . Indeed, in a fully commuted matrix element every factor of  $\Delta$  arises from an occurrence of  $L_0$ , that is, from a contraction of a positive mode of  $L_Y$  against a negative mode of  $L_{-Y'}$  of the same index through the  $2jL_0$  term of  $[L_j, L_{-j}]$ . Each contraction consumes one mode on each side, so a single term carries at most  $\min(a_Y, a_{Y'})$  factors of  $\Delta$ ; this proves the degree bound. If  $a_Y = a_{Y'}$  and the degree is attained, every mode of  $Y$  is contracted against a mode of  $Y'$  of equal index, which forces  $Y = Y'$ ; hence off-diagonal entries of equal length have degree at most  $a_Y - 1$ . For  $Y = Y'$ , the top-degree terms are the perfect matchings of equal indices: the  $m_j(Y)!$  matchings of the modes of index  $j$  each contribute  $(2j\Delta)^{m_j(Y)}$  to leading order, giving the displayed leading term.

Apply the same counting with one  $V_d$  insertion. The number of  $\Delta$ -factors of a term is again the number of positive-negative contractions producing  $L_0$ . A term that uses no Ward commutator  $[L_n, V_d(z)]$  reproduces the corresponding Shapovalov term exactly, because  $\langle \Delta | V_d(1) | \Delta \rangle = 1$ ; a term that uses at least one Ward commutator has consumed a mode without producing  $L_0$ , so its degree is strictly below  $\min(a_Y, a_{Y'})$ . It follows that, after the above rescaling, both  $G_N$  and  $\rho_N$  converge to the same invertible diagonal matrix  $D_N$ . Therefore

$$\lim_{\Delta \rightarrow \infty} F_N(c, \Delta, d) = \text{tr}(D_N^{-1} D_N) = p(N).$$

If  $e_j$  are the coefficients of  $\prod_{m \geq 1} (1 - q^m)$ , then

$$H_N = \sum_{j=0}^N e_j F_{N-j}.$$

The generating-function identity

$$\left( \sum_{j \geq 0} e_j q^j \right) \left( \sum_{k \geq 0} p(k) q^k \right) = 1$$

gives the second limit. ■

**Theorem 4.2** (*Finite pole expansion at an algebraic specialization*). Fix  $b_{\star} \in \overline{\mathbb{Q}}^{\times}$ . Then  $H_0 = 1$ , and for every  $N \geq 1$  there are unique coefficients

$$A_{\lambda, k, N}^{\star}(d) \in \mathbb{Q}(b_{\star})[d], \quad \lambda \in \Lambda_N(b_{\star}), \quad k \geq 1,$$

only finitely many of them nonzero, such that

$$H_N(c_\star, \Delta, d) = \sum_{\lambda \in \Lambda_N(b_\star)} \sum_{k \geq 1} \frac{A_{\lambda,k,N}^\star(d)}{(\Delta - \lambda)^k}.$$

The largest  $k$  with  $A_{\lambda,k,N}^\star \neq 0$  is the multiplicity of the pole of  $H_N$  at  $\lambda$ . Consequently, with

$$A_{\lambda,k}^\star(d; q) = \sum_{N \geq 1} A_{\lambda,k,N}^\star(d) q^N,$$

one has the coefficientwise locally finite expansion

$$H_\Delta^{(c_\star)}(d \mid q) = 1 + \sum_{\lambda} \sum_{k \geq 1} \frac{A_{\lambda,k}^\star(d; q)}{(\Delta - \lambda)^k},$$

where at each power of  $q$  the first sum is over the corresponding finite set  $\Lambda_N(b_\star)$ .

*Proof.* By Lemma 2.4,  $F_N = \text{tr}(G_N^{-1} \rho_N^\top)$  is rational in  $\Delta$ , and its poles are contained in the zero locus of  $\det G_N$ . The Kac determinant formula [Kac79, FF82], in the form of [IK11, Theorem 4.2 and equations (4.16)–(4.21)], gives, up to a nonzero factor independent of  $\Delta$ ,

$$\det G_N(c, \Delta) = C_N \prod_{\substack{r,s > 0 \\ rs \leq N}} (\Delta - \Delta_{r,s}(c))^{p(N-rs)}.$$

At  $c = c_\star$ , the distinct zeros belong to the finite set  $\Lambda_N(b_\star)$ . Since  $H_N$  is a finite integer linear combination of  $F_0, \dots, F_N$ , it has the same pole containment. Lemma 4.1 shows that  $H_N$  tends to zero at infinity for  $N \geq 1$ , so its partial-fraction expansion has no polynomial part and is unique.

The matrix entries belong to  $\mathbb{Q}(c_\star)[\Delta, d]$ , and every pole  $\lambda$  belongs to  $\mathbb{Q}(b_\star)$ . Laurent expansion at each pole therefore gives  $A_{\lambda,k,N}^\star \in \mathbb{Q}(b_\star)[d]$ . Coefficientwise local finiteness follows because  $rs \leq N$  leaves only finitely many labels. ■

**Theorem 4.3** (*Finite coefficientwise pole expansion*). *At  $c = 1/2$ ,  $H_0 = 1$ , and for every  $N \geq 1$  there are unique coefficients  $A_{n,k,N}(d) \in \mathbb{Q}[d]$ , only finitely many of them nonzero, such that*

$$H_N\left(\frac{1}{2}, \Delta, d\right) = \sum_{\substack{n \geq 0 \\ \exists r,s > 0: rs \leq N, |4r-3s|=n}} \sum_{k \geq 1} \frac{A_{n,k,N}(d)}{(\Delta - \lambda_n)^k}.$$

The largest  $k$  with  $A_{n,k,N} \neq 0$  is the multiplicity of the pole of  $H_N$  at  $\lambda_n$ . In particular, setting

$$A_{n,k}(d; q) = \sum_{N \geq 1} A_{n,k,N}(d) q^N$$

gives a coefficientwise locally finite expansion

$$H_\Delta^{(1/2)}(d \mid q) = 1 + \sum_{n \geq 0} \sum_{k \geq 1} \frac{A_{n,k}(d; q)}{(\Delta - \lambda_n)^k}.$$



*Proof.* Apply Theorem 4.2 at  $b_\star = b_0$ . Proposition 3.3 identifies its pole set with the stated  $\lambda_n$ 's. The specialized inverse-Shapovalov coefficient has rational coefficients, and all  $\lambda_n$  are rational, so uniqueness of partial fractions improves the coefficient field from  $\mathbb{Q}(b_0)$  to  $\mathbb{Q}$ . Local finiteness is Corollary 3.4.  $\blacksquare$

The conceptual result can now be stated independently of the Ising calculations. Its collision moments and finite precision budget are defined in Section 7, after the generic residue data; the proof is given there as well.

**Theorem 4.4** (*Finite algebraic confluence at fixed  $b_\star$* ). Fix  $b_\star \in \overline{\mathbb{Q}}^\times$ . For every  $N \geq 1$ ,

$$H_N(c_\star, \Delta, d) = \sum_{\lambda \in \Lambda_N(b_\star)} \sum_{k=1}^{M_{\lambda,N}^\star} \frac{\tilde{A}_{\lambda,k,N}^\star(d)}{(\Delta - \lambda)^k}. \quad (4.1)$$

The sums and every constant-term calculation are finite,  $\tilde{A}_{\lambda,k,N}^\star(d) \in \mathbb{Q}(b_\star)[d]$ , and the level-decreasing recursion of Definitions 7.1–7.3 terminates. Its backward precision rule supplies a finite truncation order for every intermediate Laurent germ. Moreover,

$$\tilde{A}_{\lambda,k,N}^\star = A_{\lambda,k,N}^\star, \quad m_{\lambda,N}^\star \leq M_{\lambda,N}^\star \leq \widehat{M}_{\lambda,N}^\star,$$

where  $m_{\lambda,N}^\star$  is the exact pole multiplicity. Consequently the resulting locally finite series agrees coefficient by coefficient in  $q$  with the inverse-Shapovalov definition of the meromorphic Verma-module torus one-point block.

## 5. Local Smith exponents, transverse collisions, and the first residues

**Lemma 5.1** (*Pole order from local invariant factors*). Let  $R = \mathbb{C}[[x]]$ , and let  $G(x) \in M_m(R)$  have nonzero determinant. Suppose the Smith exponents of  $G$  are  $0 \leq e_1 \leq \dots \leq e_m$ . Then

$$\text{ord}_x \det G = \sum_{i=1}^m e_i, \quad \dim \ker G(0) = \#\{i : e_i > 0\},$$

and the pole order of  $G(x)^{-1}$  is  $e_m$ . In particular, if

$$\text{ord}_x \det G = \dim \ker G(0) = k,$$

then every positive Smith exponent equals 1, and  $G^{-1}$  has at most a simple pole.

*Proof.* Since  $R$  is a discrete valuation ring, there are  $U, V \in \text{GL}_m(R)$  such that

$$UGV = \text{diag}(x^{e_1}, \dots, x^{e_m}).$$

Taking determinants gives the first identity. Reducing modulo  $x$  gives the second. Inverting the displayed Smith form gives pole order  $e_m$ . If the sum of the  $k$  positive integers  $e_i$  is  $k$ , then each of them is 1.  $\blacksquare$

**Proposition 5.2** (*Transverse inverse residue*). *Let  $V$  be finite dimensional, let  $G(x): V \rightarrow V^*$  be a symmetric matrix over  $\mathbb{C}[[x]]$  with nonzero determinant, and put  $K = \ker G(0)$ . Let  $\iota: K \hookrightarrow V$  be the inclusion and define the crossing form*

$$\beta(u, v) = u^\top G'(0)v, \quad u, v \in K.$$

*If  $\beta$  is nondegenerate, then  $G(x)^{-1}$  has only a simple pole and*

$$\operatorname{Res}_{x=0} G(x)^{-1} = \iota \beta^{-1} \iota^*.$$

*Moreover,  $\beta$  is nondegenerate if and only if*

$$\operatorname{ord}_x \det G = \dim K.$$

*Proof.* Choose a complement  $C$  of  $K$ . Since  $K$  is the radical of the symmetric form  $G(0)$ , its restriction  $A$  to  $C$  is nondegenerate. In the decomposition  $V = K \oplus C$ , one therefore has

$$G(x) = \begin{pmatrix} xB + O(x^2) & xD + O(x^2) \\ xD^\top + O(x^2) & A + O(x) \end{pmatrix},$$

where  $B$  is the matrix of  $\beta$ . The Schur complement of the lower-right block is

$$S(x) = xB + O(x^2).$$

If  $B$  is invertible, block inversion gives

$$G(x)^{-1} = \frac{1}{x} \begin{pmatrix} B^{-1} & 0 \\ 0 & 0 \end{pmatrix} + O(1),$$

which is the coordinate expression of the asserted residue. Also

$$\det G(x) = \det(A + O(x)) \det S(x).$$

Every entry of  $S$  is divisible by  $x$ , so its determinant is divisible by  $x^{\dim K}$ , with equality of orders precisely when  $\det B \neq 0$ . This proves the final equivalence.  $\blacksquare$

**Proposition 5.3** (*Two-step inverse principal part*). *Let*

$$G(x) = G_0 + xG_1 + x^2G_2 + O(x^3)$$

*be as in Proposition 5.2, and put*

$$K_1 = \ker G_0, \quad \beta_1(u, v) = u^\top G_1 v, \quad K_2 = \ker(\beta_1: K_1 \rightarrow K_1^*).$$

*For  $v \in K_2$ , choose  $v_1 \in V$  satisfying*

$$G_0 v_1 = -G_1 v$$

*and define*

$$\beta_2(u, v) = u^\top (G_2 v + G_1 v_1), \quad u, v \in K_2.$$

Then  $v_1$  exists,  $\beta_2$  is independent of its choice, and  $\beta_2$  is symmetric. Without any nondegeneracy assumption,

$$\#\{i : e_i \geq 1\} = \dim K_1, \quad \#\{i : e_i \geq 2\} = \dim K_2.$$

$\beta_2$  is nondegenerate if and only if all Smith exponents are at most 2. In that case,

$$[x^{-2}]G(x)^{-1} = \iota_2 \beta_2^{-1} \iota_2^*,$$

where  $\iota_2 : K_2 \hookrightarrow V$ .

*Proof.* Because  $G_0$  is symmetric,  $\text{im } G_0$  is the annihilator of  $K_1$ . For  $v \in K_2$  and  $u \in K_1$ ,

$$u^\top G_1 v = \beta_1(u, v) = 0.$$

Thus  $G_1 v \in \text{im } G_0$ , proving the existence of  $v_1$ . Two choices differ by an element  $k \in K_1$ , and  $u^\top G_1 k = \beta_1(u, k) = 0$  for  $u \in K_2$ , proving choice-independence.

Choose a complement to  $K_1$  as in the proof of Proposition 5.2, and first take the Schur complement of its nondegenerate block. This produces a symmetric matrix on  $K_1$  of the form

$$S(x) = xB_1 + x^2B_2 + O(x^3),$$

where  $B_1$  represents  $\beta_1$ , and the restriction of  $B_2$  to  $\ker B_1 = K_2$  is the displayed form  $\beta_2$ . This also proves its symmetry. The positive Smith exponents of  $G$  are the Smith exponents of  $S$ . Dividing  $S$  by  $x$  and reducing modulo  $x$  gives  $B_1$ . The Smith normal form, first for  $G$  and then for  $x^{-1}S$ , therefore gives

$$\#\{i : e_i \geq 1\} = \dim K_1, \quad \#\{i : e_i \geq 2\} = \dim \ker B_1 = \dim K_2.$$

This count does not require the second crossing form to be nondegenerate.

Now choose a complement  $E$  to  $K_2$  in  $K_1$ . The restriction of  $B_1$  to  $E$  is nondegenerate, and a second Schur complement gives

$$S(x) \sim \begin{pmatrix} x^2\beta_2 + O(x^3) & 0 \\ 0 & xB_1|_E + O(x^2) \end{pmatrix}.$$

The second block form shows that  $\beta_2$  is nondegenerate precisely when no Smith exponent exceeds 2. In that case block inversion yields the asserted  $x^{-2}$  coefficient.  $\blacksquare$

**Corollary 5.4** (*Determinant multiplicity does not determine block pole order*). *At a collided Kac weight, the multiplicity of the corresponding factor in  $\det G_N$  is only the sum of the positive local Smith exponents. The inverse Shapovalov pole order is their maximum.*

*Proof.* Apply Lemma 5.1 with  $x = \Delta - \lambda_n$ .  $\blacksquare$

**Lemma 5.5** (*Torus form on a singular submodule*). *Let  $\chi \in M(c, h)$  be a singular vector of level  $\ell$ , normalized by a fixed PBW coefficient, and let*

$$P_\chi(d) = \langle \chi | V_d(1) | \chi \rangle.$$

*For partitions  $Y, Y'$  of the same integer  $m$ ,*

$$\langle \chi | L_Y V_d(1) L_{-Y'} | \chi \rangle = P_\chi(d) \rho_m(c, h + \ell, d)_{Y, Y'}.$$

*Proof.* The assignment

$$L_{-Y}|h + \ell\rangle \longmapsto L_{-Y}\chi$$

is a homomorphism from  $M(c, h + \ell)$  to  $M(c, h)$ , because  $\chi$  is singular and has  $L_0$ -weight  $h + \ell$ . Pull the torus three-point form on  $M(c, h)$  back along this homomorphism in both module arguments. Commuting one Virasoro mode at a time through  $V_d(z)$  shows that the pulled-back form obeys the same Ward recursion as  $\rho_m(c, h + \ell, d)$ . That recursion terminates at the highest-weight matrix element and therefore determines every descendant matrix element uniquely. The pulled-back highest-weight value is  $P_\chi(d)$ , whereas the normalization  $\langle h + \ell | V_d(1) | h + \ell \rangle = 1$  gives highest-weight value 1 for  $\rho_m$ . The displayed identity follows.  $\blacksquare$

**Lemma 5.6** (*First crossing form on a singular submodule*). *Fix  $c$  and let  $\chi = P|h_0\rangle \in M(c, h_0)$  be singular of level  $\ell$ , where the PBW operator  $P \in U(\text{Vir}_-)$  is held fixed as  $h$  varies. Set*

$$b_\chi = \left. \frac{d}{dh} \langle h | \omega(P) P | h \rangle \right|_{h=h_0},$$

where  $\omega(L_n) = L_{-n}$ . For partitions  $Y, Y'$  of  $m$ ,

$$\left. \frac{d}{dh} \langle h | \omega(P) L_Y L_{-Y'} P | h \rangle \right|_{h=h_0} = b_\chi G_m(c, h_0 + \ell)_{Y, Y'}.$$

*Proof.* Write  $S_h(u, v) = \langle h | \omega(u) v | h \rangle$  for the Shapovalov form of  $M(c, h)$ , and for  $u \in U(\text{Vir}_-)$  put  $\tilde{u} = uP \in U(\text{Vir}_-)$ . The derivative on the left defines a bilinear form

$$C(u, v) = \left. \frac{d}{dh} S_h(\tilde{u}, \tilde{v}) \right|_{h=h_0}$$

on the PBW space of  $M(c, h_0 + \ell)$ . At  $h = h_0$ , the map

$$j_\chi: M(c, h_0 + \ell) \longrightarrow M(c, h_0), \quad u|h_0 + \ell\rangle \longmapsto u\chi,$$

is a module homomorphism, and its image lies in the radical of the Shapovalov form.

We claim that  $C$  is contravariant for the module structure of  $M(c, h_0 + \ell)$ . Contravariance in negative modes,  $C(L_{-n}u, v) = C(u, \omega(L_{-n})v)$ , holds at every  $h$  because the Shapovalov form itself is contravariant and  $\widetilde{L_{-n}u} = L_{-n}\tilde{u}$ . For a positive mode, fix  $n > 0$  and  $v \in U(\text{Vir}_-)$ , and let  $w_h \in U(\text{Vir}_-)$  be defined by PBW reduction of  $L_n v$  in  $M(c, h_0 + \ell)$ , so that  $L_n v | h_0 + \ell \rangle = w_{h_0} | h_0 + \ell \rangle$ . In  $M(c, h)$  set

$$D(h) = L_n \tilde{v} | h \rangle - \widetilde{w_{h_0}} | h \rangle.$$

Each coefficient of  $D(h)$  in the PBW basis is polynomial in  $h$ , and  $D(h_0) = 0$  because  $\chi = P|h_0\rangle$  is singular and  $j_\chi$  intertwines the actions at  $h = h_0$ . Hence  $D(h) = (h - h_0)\tilde{D}(h)$  with  $\tilde{D}$  polynomial in  $h$ . Therefore

$$\left. \frac{d}{dh} \right|_{h_0} S_h(\tilde{u}, L_n \tilde{v}) = \left. \frac{d}{dh} \right|_{h_0} S_h(\tilde{u}, \widetilde{w_{h_0}}) + S_{h_0}(\tilde{u}, \tilde{D}(h_0)).$$

The last term vanishes because  $\tilde{u}|h_0\rangle = j_\chi(u|h_0 + \ell)\rangle$  lies in the radical of  $S_{h_0}$ . Since  $S_h(\tilde{u}, L_n\tilde{v}) = S_h(\widetilde{L_{-n}u}, \tilde{v})$ , this proves  $C(L_{-n}u, v) = C(u, w_{h_0})$ , that is, contravariance in positive modes.

A contravariant form on a Verma module is uniquely determined by its highest-weight value, because PBW commutation recursively reduces every matrix element to that value. Here the highest-weight value is  $b_\chi$ , so  $C = b_\chi G_m(c, h_0 + \ell)$ .  $\blacksquare$

**Proposition 5.7** (*Earliest Ising collision*). *The first repeated Ising Kac weight occurs at level 3:*

$$\Delta_{2,1} = \Delta_{1,3} = \frac{1}{2}, \quad \ell_{2,1} = 2, \quad \ell_{1,3} = 3.$$

In  $M(1/2, 1/2)$ , the vectors

$$\xi_2 = \left( L_{-1}^2 - \frac{4}{3}L_{-2} \right) \left| \frac{1}{2} \right\rangle$$

and

$$\xi_3 = \left( L_{-1}^3 - 3L_{-2}L_{-1} + \frac{3}{4}L_{-3} \right) \left| \frac{1}{2} \right\rangle$$

are singular. At level 3,

$$\ker G_3(1/2, 1/2) = \text{span}\{L_{-1}\xi_2, \xi_3\},$$

and the local Smith exponents are  $(0, 1, 1)$ . With the  $L_{-1}^\ell$ -coefficient normalized to 1, their crossing scalars are

$$b_{\xi_2} = \frac{28}{9}, \quad b_{\xi_3} = -\frac{105}{8},$$

and the level-three crossing form in the ordered basis  $(L_{-1}\xi_2, \xi_3)$  is

$$\begin{pmatrix} 140/9 & 0 \\ 0 & -105/8 \end{pmatrix} = \begin{pmatrix} \frac{28}{9}G_1(1/2, 5/2) & 0 \\ 0 & -105/8 \end{pmatrix}.$$

Moreover, the diagonal torus matrix elements are

$$E_2(d) := \langle \xi_2 | V_d(1) | \xi_2 \rangle = \frac{d(d-1)(3d-14)(3d-5)}{9},$$

and

$$E_3(d) := \langle \xi_3 | V_d(1) | \xi_3 \rangle = \frac{d(d-1)(2d-15)(2d-7)(2d-5)(2d-1)}{16}.$$

*Proof.* The collision statement follows immediately from  $\Delta_{r,s} = \lambda_{|4r-3s|}$ : the labels of levels at most 2 have distinct values of  $|4r-3s|$ , whereas both displayed labels have value 5. Direct commutation with  $L_1, L_2, L_3$  proves that the two vectors are singular. In the ordered basis  $(L_{-3}, L_{-2}L_{-1}, L_{-1}^3)|\Delta\rangle$ ,

$$\det G_3(1/2, \Delta) = 12\Delta^2(2\Delta-1)^2(3\Delta-5)(16\Delta-1).$$

At  $\Delta = 1/2$  the matrix has rank one, and

$$L_{-1}\xi_2 = \left( -\frac{4}{3}L_{-3} - \frac{4}{3}L_{-2}L_{-1} + L_{-1}^3 \right) \left| \frac{1}{2} \right\rangle$$

together with  $\xi_3$  spans its radical. The determinant valuation and nullity are both two, so Lemma 5.1 gives the Smith profile. Finally,

$$b_{\xi_j} = \left. \frac{d}{d\Delta} \langle \xi_j | \xi_j \rangle \right|_{\Delta=1/2}$$

and direct multiplication by  $\partial_\Delta G_3|_{\Delta=1/2}$  gives the displayed level-three crossing matrix. The Ward recursion of Lemma 2.4 gives the two displayed torus polynomials. ■

**Theorem 5.8** (*First fixed- $c$  recursive residue*). *At  $c = 1/2$ , the earliest collided coefficient has the simple principal part*

$$\text{Res}_{\Delta=1/2} H_3(1/2, \Delta, d) = \frac{9}{28} E_2(d) H_1(1/2, 5/2, d) - \frac{8}{105} E_3(d).$$

Every quantity on the right is evaluated directly at  $c = 1/2$ .

*Proof.* The block-diagonal crossing form in Proposition 5.7 has highest-weight inverse scalars  $9/28$  and  $-8/105$ . Proposition 5.2 and Lemma 5.5 therefore give

$$\text{Res}_{\Delta=1/2} F_2 = \frac{9}{28} E_2, \quad \text{Res}_{\Delta=1/2} F_3 = \frac{9}{28} E_2 F_1(1/2, 5/2, d) - \frac{8}{105} E_3.$$

The Euler product gives  $H_3 = F_3 - F_2 - F_1$ ; since  $G_1(1/2, \Delta) = (2\Delta)$ , the coefficient  $F_1$  is regular at  $\Delta = \frac{1}{2}$ , so  $\text{Res}_{\Delta=1/2} H_3 = \text{Res}_{\Delta=1/2} (F_3 - F_2)$ . The result follows from  $H_1 = F_1 - F_0$ . ■

**Proposition 5.9** (*First spin-module collision*). *At level 4, the labels  $(1, 2)$  and  $(2, 2)$  collide at  $\Delta = 1/16$ . In the PBW basis*

$$(L_{-4}, L_{-3}L_{-1}, L_{-2}^2, L_{-2}L_{-1}^2, L_{-1}^4)|\Delta\rangle,$$

the Shapovalov determinant at  $c = 1/2$  is

$$12\Delta^3(2\Delta - 7)(2\Delta - 1)^3(3\Delta - 5)(16\Delta - 21)(16\Delta - 1)^3.$$

At  $\Delta = 1/16$ , the local Smith exponents are

$$(0, 0, 1, 1, 1).$$

Consequently,  $G_4(1/2, \Delta)^{-1}$ ,  $F_4(1/2, \Delta, d)$ , and  $H_4(1/2, \Delta, d)$  have at most a simple pole at  $\Delta = 1/16$ .

*Proof.* Direct Virasoro commutation gives the  $5 \times 5$  Gram matrix  $G_4(1/2, \Delta)$  recorded in Appendix B. Its determinant is the polynomial displayed in the statement. At  $\Delta = 1/16$ , the upper-left  $2 \times 2$  minor has determinant 2, while the following three independent vectors lie in the kernel:

$$\begin{pmatrix} -9/16 \\ -3/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -27/64 \\ -9/8 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -465/256 \\ -75/32 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Thus the specialized matrix has rank 2 and nullity 3. The determinant has order 3 at  $1/16$ , so Lemma 5.1 gives the stated Smith exponents and inverse pole bound. The definitions of  $F_4$  and  $H_4$  then give the same bound for the block coefficients. ■

**Proposition 5.10** (*Singular data at the first spin collision*). *Put  $h_0 = 1/16$ . The following vectors in  $M(1/2, h_0)$  are singular:*

$$\chi_2 = \left( L_{-1}^2 - \frac{3}{4}L_{-2} \right) |h_0\rangle$$

and

$$\chi_4 = \left( L_{-1}^4 - \frac{25}{6}L_{-2}L_{-1}^2 + \frac{49}{144}L_{-2}^2 + \frac{11}{6}L_{-3}L_{-1} - \frac{1}{4}L_{-4} \right) |h_0\rangle.$$

At level 4,

$$\ker G_4(1/2, h_0) = \text{span}\{L_{-2}\chi_2, L_{-1}^2\chi_2, \chi_4\}.$$

In this ordered basis the crossing form of Proposition 5.2, with  $x = \Delta - h_0$ , is

$$\beta_4 = \begin{pmatrix} -119/8 & -693/32 & 0 \\ -693/32 & -9471/128 & 0 \\ 0 & 0 & 13475/648 \end{pmatrix} = \begin{pmatrix} -\frac{7}{4}G_2(1/2, 33/16) & 0 \\ 0 & 13475/648 \end{pmatrix}.$$

Finally, the two highest-weight torus matrix elements are

$$P_2(d) := \langle \chi_2 | V_d(1) | \chi_2 \rangle = \frac{d(d-1)(2d-7)(2d-1)}{4}$$

and

$$\begin{aligned} P_4(d) &:= \langle \chi_4 | V_d(1) | \chi_4 \rangle \\ &= \frac{d(d-1)(2d-7)(2d-1)(3d-14)(3d-5)(6d-55)(6d-1)}{1296}. \end{aligned}$$

*Proof.* Apply  $L_1, \dots, L_\ell$  to the displayed level- $\ell$  vector and use the Virasoro commutator. Collecting coefficients in the PBW bases at levels  $\ell-1, \dots, 0$  gives zero for  $\chi_2$  and for  $\chi_4$ ; modes larger than  $\ell$  vanish by degree. Thus both vectors are singular.

In the level-four PBW basis of Proposition 5.9, the three asserted kernel vectors are the columns

$$\begin{pmatrix} 0 \\ 0 \\ -3/4 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -3/2 \\ -3/2 \\ 0 \\ -3/4 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1/4 \\ 11/6 \\ 49/144 \\ -25/6 \\ 1 \end{pmatrix}.$$

Multiplication by the matrix  $G_4(1/2, h_0)$  of Appendix B gives zero. They are independent, and Proposition 5.9 gives nullity three, proving the kernel identity.

Multiplying these columns on both sides of  $\partial_\Delta G_4(1/2, \Delta)|_{\Delta=h_0}$  gives the displayed crossing matrix. Direct level-two commutation gives

$$G_2(1/2, 33/16) = \begin{pmatrix} 17/2 & 99/8 \\ 99/8 & 1353/32 \end{pmatrix},$$

which verifies the second expression for  $\beta_4$ . Finally, repeated use of the primary-field Ward commutator in the proof of Lemma 2.4, followed by collection and factorization over  $\mathbb{Q}[d]$ , gives the displayed  $P_2$  and  $P_4$ . ■

**Theorem 5.11** (*Spin-sector fixed- $c$  recursive principal part*). *At  $c = 1/2$ , the level-four reduced coefficient satisfies the finite fixed-central-charge recursion*

$$\operatorname{Res}_{\Delta=1/16} H_4(1/2, \Delta, d) = -\frac{4}{7}P_2(d)H_2(1/2, 33/16, d) + \frac{648}{13475}P_4(d).$$

Equivalently, this residue is

$$\frac{d(d-1)(2d-7)(2d-1)}{40425} (248d^4 - 5752d^3 + 25274d^2 - 33630d + 5775).$$

Every quantity on the right is evaluated directly at  $c = 1/2$ .

*Proof.* At levels 2, 3, and 4, the radicals at  $\Delta = h_0$  are respectively

$$\operatorname{span}\{\chi_2\}, \quad \operatorname{span}\{L_{-1}\chi_2\}, \quad \operatorname{span}\{L_{-2}\chi_2, L_{-1}^2\chi_2, \chi_4\}.$$

The crossing forms on the first two spaces are

$$-\frac{7}{4}, \quad -\frac{7}{4}G_1(1/2, 33/16),$$

and the third is the form in Proposition 5.10. These identities follow by differentiating the level-two, level-three, and level-four Gram matrices; all three crossing forms are nondegenerate. Apply Proposition 5.2, then use Lemma 5.5 on each singular summand. Taking the trace against the corresponding torus matrix gives

$$\begin{aligned} \operatorname{Res}_{\Delta=h_0} F_2 &= -\frac{4}{7}P_2, \\ \operatorname{Res}_{\Delta=h_0} F_3 &= -\frac{4}{7}P_2F_1(1/2, 33/16, d), \\ \operatorname{Res}_{\Delta=h_0} F_4 &= -\frac{4}{7}P_2F_2(1/2, 33/16, d) + \frac{648}{13475}P_4. \end{aligned}$$

The Euler product has coefficients 1, -1, -1, 0, 0 through degree four. Hence  $H_4 = F_4 - F_3 - F_2$ , while  $H_2 = F_2 - F_1 - F_0$ . Substitution gives the recursive formula. Inserting the explicit polynomials from Proposition 5.10 and the level-two coefficient from the inverse Shapovalov matrix gives the second displayed expression. ■

## 6. Generic residue data

**Definition 6.1** (Generic residue data used for algebraic confluence). Use the functions  $c(b)$  and  $h_{r,s}(b)$  of Definition 3.1. Put  $\Lambda^2 = (b + b^{-1})^2 - 4d$ , and define

$$\begin{aligned} \mathcal{P}_{r,s}(b, d) &= \prod_{\substack{1 \leq k \leq 2r-1 \\ k \text{ odd}}} \prod_{\substack{1 \leq l \leq 2s-1 \\ l \text{ odd}}} \frac{\Lambda^2 - (kb + lb^{-1})^2}{4} \\ &\quad \times \frac{\Lambda^2 - (kb - lb^{-1})^2}{4}, \end{aligned}$$



and

$$\mathcal{A}_{r,s}(b) = \frac{1}{2} \prod_{\substack{1-r \leq p \leq r, \ 1-s \leq q \leq s \\ (p,q) \neq (0,0), (r,s)}} (pb + qb^{-1})^{-1}, \quad \mathcal{R}_{r,s}(b, d) = \mathcal{A}_{r,s}(b) \mathcal{P}_{r,s}(b, d).$$

For generic  $b$ ,  $\mathcal{R}_{r,s}$  is the residue coefficient in the eta-reduced torus recursion, stated as Proposition 6.4 below.

*Remark 6.2* (Source, normalization, and imported identities). The recursion was conjectured in [Pog09]. In [HJS10, Section 3], equation (7) expresses the residue through the inverse norm derivative  $A_{r,s}(c)$ , the paragraph immediately following equation (7) gives its product for the normalization in which the singular vector has  $L_{-1}^{rs}$ -coefficient 1, equation (8) evaluates the product of the two torus fusion polynomials, and equation (9) is the eta-reduced recursion. Definition 6.1 is exactly this chain after writing the four linear factors of equation (8) as the two quadratic factors in  $\mathcal{P}_{r,s}$ , without choosing a square root of  $\Lambda^2$ .

Two normalization identities at generic  $b$  are imported from that derivation rather than rederived here. First, with  $\chi_{r,s}(b)$  the normalized singular vector and  $b_\chi$  the crossing scalar of Lemma 5.6,

$$b_{\chi_{r,s}(b)} = \mathcal{A}_{r,s}(b)^{-1}; \quad (6.1)$$

this is equation (7) and the displayed product following it, translated from the inverse norm derivative to the crossing-form convention of Lemma 5.6. Second,

$$\langle \chi_{r,s}(b) | V_d(1) | \chi_{r,s}(b) \rangle = \mathcal{P}_{r,s}(b, d); \quad (6.2)$$

this is the factorization displayed between equations (7) and (8), followed by the explicit fusion-polynomial product in equation (8), and is *not* a one-line consequence of the Ward recursion. Both sides of each identity are rational in  $b$ , so the identities continue to every specialization at which the normalized singular vector is regular. This fixes the normalization used below, rather than merely identifying the polynomials up to a scalar. Both identities are verified in exact arithmetic by the audit scripts of Remark 9.6: (6.2) directly at  $b_0$  for the labels  $(2, 1), (1, 3), (1, 5), (4, 1), (1, 2), (2, 2)$ , at a generic sample point for  $(2, 1), (1, 2)$ , and, for  $(4, 2), (1, 6)$ , through the level-eight and level-six residues checked by `tools/audit_grade10_direct.py`; and (6.1) at  $b_0$  for the labels  $(2, 1), (1, 3), (1, 2), (2, 2)$  through the crossing scalars of Section 5.

**Lemma 6.3** (*Distinct weights and no shifted self-poles at generic  $b$* ). For positive labels  $(r, s)$  and  $(u, v)$ ,

$$h_{r,s}(b) + rs - h_{u,v}(b)$$

is not the zero rational function of  $b$ . Moreover, for distinct positive labels  $(r, s) \neq (u, v)$ , the difference  $h_{r,s}(b) - h_{u,v}(b)$  is not the zero rational function.

*Proof.* The Kac formula gives the identity

$$h_{r,s}(b) + rs = h_{r,-s}(b).$$

If this were identically equal to  $h_{u,v}(b)$ , then

$$(rb - sb^{-1})^2 = (ub + vb^{-1})^2$$

in  $\mathbb{C}(b)$ . Since this is an integral domain,

$$rb - sb^{-1} = \pm(ub + vb^{-1}).$$

Comparison of the coefficients of  $b$  and  $b^{-1}$  contradicts  $r, s, u, v > 0$ . For the second assertion, the same argument gives  $rb + sb^{-1} = \pm(ub + vb^{-1})$ , and positivity forces  $(r, s) = (u, v)$ .  $\blacksquare$

**Proposition 6.4** (*Generic eta-reduced recursion*). *Fix  $N \geq 1$ . For all  $b$  outside a finite set of algebraic values,*

$$H_N(c(b), \Delta, d) = \sum_{\substack{a=(r,s) \in \mathbb{Z}_{>0}^2 \\ \ell_a = rs \leq N}} \frac{\mathcal{R}_{r,s}(b, d)}{\Delta - h_{r,s}(b)} H_{N-\ell_a}(c(b), h_{r,s}(b) + rs, d)$$

as an identity of rational functions of  $\Delta$ , polynomial in  $d$ .

*Proof.* Given the two imported identities of Remark 6.2, the proposition follows from the results of Sections 4 and 5. By Lemma 6.3, outside finitely many values of  $b$  the weights  $h_{r,s}(b)$  with  $rs \leq N$  are pairwise distinct, and no shifted weight  $z_{r,s} = h_{r,s}(b) + rs$  is a degenerate weight of level at most  $N$ .

Fix such a  $b$ , write  $c = c(b)$ , and fix a label  $a = (r, s)$  with  $rs \leq N$ . By distinctness,  $h_{r,s}(b)$  lies on no other Kac divisor of level at most  $N$ , so the level- $N$  Kac determinant has valuation exactly  $p(N - rs)$  at  $\Delta = h_{r,s}(b)$ . The level- $N$  image of the injective singular embedding  $j_{\chi_{r,s}(b)}$  is a  $p(N - rs)$ -dimensional subspace of the kernel of the specialized Gram matrix. By Lemma 5.1 the nullity is at most the valuation, so the kernel is exactly that image, and every positive local Smith exponent equals 1; hence Proposition 5.2 applies. By Lemma 5.6, the crossing form on this kernel is  $b_\chi G_{N-rs}(c, z_{r,s})$ , which is nondegenerate because  $z_{r,s}$  is not a degenerate weight and, by (6.1),  $b_\chi^{-1} = \mathcal{A}_{r,s}(b) \neq 0$ . Taking the trace of the residue of  $G_N^{-1}$  against  $\rho_N$  and factoring the pulled-back torus form by Lemma 5.5 and (6.2) gives

$$\operatorname{Res}_{\Delta=h_{r,s}(b)} F_N(c, \Delta, d) = \mathcal{A}_{r,s}(b) \mathcal{P}_{r,s}(b, d) F_{N-rs}(c, z_{r,s}, d) = \mathcal{R}_{r,s}(b, d) F_{N-rs}(c, z_{r,s}, d).$$

The difference between  $F_N$  and the sum of these principal parts has no finite pole, hence is a polynomial in  $\Delta$ ; it is bounded as  $\Delta \rightarrow \infty$  by Lemma 4.1, hence equals the constant  $p(N)$ :

$$F_N(c, \Delta, d) = p(N) + \sum_{\ell_a \leq N} \frac{\mathcal{R}_a(b, d)}{\Delta - h_a(b)} F_{N-\ell_a}(c, z_a, d).$$

Finally, multiply by the Euler coefficients  $e_j$  and sum. The generating identity in the proof of Lemma 4.1 gives  $\sum_j e_j p(N - j) = \delta_{N,0}$ , while the residue sums reassemble into shifted

$H$ -coefficients:

$$H_N(c, \Delta, d) = \delta_{N,0} + \sum_{\ell_a \leq N} \frac{\mathcal{R}_a(b, d)}{\Delta - h_a(b)} H_{N-\ell_a}(c, z_a, d).$$

For  $N \geq 1$  this is the displayed recursion. ■

## 7. All-order confluent recursion

**Definition 7.1** (Shifted Laurent recursion). Fix  $b_\star \in \overline{\mathbb{Q}}^\times$ , put  $c_\star = c(b_\star)$ , let  $t = b - b_\star$ , and work in

$$K_\star((t)), \quad K_\star = \mathbb{Q}(b_\star, d),$$

where  $d$  is transcendental. For a positive Kac label  $a = (r, s)$ , write

$$\ell_a = rs, \quad h_a(t) = h_{r,s}(b_\star + t), \quad R_a(t) = \mathcal{R}_{r,s}(b_\star + t, d), \quad z_a(t) = h_a(t) + \ell_a.$$

Define shifted Laurent coefficients recursively by

$$S_{a,0}(t) = 1$$

and, for  $M \geq 1$ ,

$$S_{a,M}(t) = \sum_{\substack{e=(u,v) \in \mathbb{Z}_{>0}^2 \\ \ell_e = uv \leq M}} \frac{R_e(t)}{z_a(t) - h_e(t)} S_{e,M-\ell_e}(t).$$

Every denominator in this formula is a nonzero rational function of  $b$  by Lemma 6.3, so each summand has a well-defined Laurent expansion, and the recursion terminates because its second index strictly decreases.

**Lemma 7.2** (*A priori shifted valuation bound*). For a positive label  $e$ , and for two positive labels  $a, e$ , set

$$\rho_e = \max\{0, -\text{ord}_t R_e(t)\}, \quad \sigma_{a,e} = \max\{0, \text{ord}_t(z_a(t) - h_e(t))\}.$$

Define nonnegative integers  $W_{a,M}$  recursively by

$$W_{a,0} = 0,$$

and, for  $M \geq 1$ ,

$$W_{a,M} = \max(\{0\} \cup \{\rho_e + \sigma_{a,e} + W_{e,M-\ell_e} : \ell_e \leq M\}).$$

Then

$$\text{ord}_t S_{a,M}(t) \geq -W_{a,M}.$$

In particular,  $W_{a,M}$  is an a priori pole bound computed solely from the finite recursion tree and the valuations of its explicit residue and denominator factors.

*Proof.* The assertion is clear for  $M = 0$ . If it holds at lower levels, the  $e$ -summand in Definition 7.1 has valuation at least

$$-\rho_e - \sigma_{a,e} - W_{e,M-\ell_e}.$$

The valuation of a finite sum is at least the minimum of the summand valuations. Taking the displayed maximum proves the induction step. All sets involved are finite because  $\ell_e \leq M$ .  $\blacksquare$

**Definition 7.3** (Collision moments). For  $N \geq 1$ ,  $\lambda \in \Lambda_N(b_*)$ , and  $a \in \mathcal{C}_{\lambda,N}^*$ , define

$$B_{a,N}(t) = R_a(t)S_{a,N-\ell_a}(t), \quad \delta_a(t) = h_a(t) - \lambda.$$

If  $\delta_a \not\equiv 0$  and  $B_{a,N} \not\equiv 0$ , put

$$\varepsilon_a = \text{ord}_t \delta_a, \quad v_{a,N} = \max\{0, -\text{ord}_t B_{a,N}\}.$$

If  $\delta_a \not\equiv 0$  and  $B_{a,N} \equiv 0$ , put  $\varepsilon_a = \text{ord}_t \delta_a$  and  $v_{a,N} = 0$ . If  $\delta_a \equiv 0$ , set  $\varepsilon_a = \infty$  and  $v_{a,N} = 0$ . Define also the a priori numerator bound

$$\widehat{v}_{a,N} = \rho_a + W_{a,N-\ell_a}.$$

Lemma 7.2 gives

$$v_{a,N} \leq \widehat{v}_{a,N}.$$

Define the exact finite structural pole bound

$$M_{\lambda,N}^* = \max_{a \in \mathcal{C}_{\lambda,N}^*} \begin{cases} 1 + \lfloor v_{a,N}/\varepsilon_a \rfloor, & \varepsilon_a < \infty, \\ 1, & \varepsilon_a = \infty, \end{cases}$$

and its a priori majorant

$$\widehat{M}_{\lambda,N}^* = \max_{a \in \mathcal{C}_{\lambda,N}^*} \begin{cases} 1 + \lfloor \widehat{v}_{a,N}/\varepsilon_a \rfloor, & \varepsilon_a < \infty, \\ 1, & \varepsilon_a = \infty. \end{cases}$$

Thus  $M_{\lambda,N}^* \leq \widehat{M}_{\lambda,N}^*$ . The former can be sharpened after the exact rational germs have been computed, while the latter bounds the number of moments before those cancellations are simplified. For  $1 \leq k \leq M_{\lambda,N}^*$ , define the *collision moments*

$$\widetilde{A}_{\lambda,k,N}^*(d) = \text{CT}_t \sum_{a \in \mathcal{C}_{\lambda,N}^*} B_{a,N}(t) \delta_a(t)^{k-1}. \quad (7.1)$$

Here  $\text{CT}_t$  denotes the coefficient of  $t^0$  in a Laurent series. Put

$$m_{\lambda,N}^* = \max(\{k : \widetilde{A}_{\lambda,k,N}^* \neq 0\} \cup \{0\}).$$

It satisfies  $m_{\lambda,N}^* \leq M_{\lambda,N}^*$ , and Theorem 4.4 identifies it with the exact multiplicity of the pole of  $H_N$  at  $\lambda$ .

**Lemma 7.4** (*Backward Laurent-precision budget*). Put

$$D_{a,e}(t) = \frac{R_e(t)}{z_a(t) - h_e(t)}, \quad p_{a,e} = \rho_e + \sigma_{a,e}.$$

Suppose  $S_{a,M}$  is required through the coefficient of  $t^U$ . For its  $e$ -summand it is sufficient to know

$$D_{a,e} \text{ through } t^{U+W_{e,M}-\ell_e}, \quad S_{e,M-\ell_e} \text{ through } t^{U+p_{a,e}}. \quad (7.2)$$

Iterating these requests down the strictly level-decreasing recursion tree produces a finite a priori upper precision at every node.

In particular, to compute every collision moment at level  $N$ , it is sufficient for each  $a$  to compute  $R_a$  through  $t^{W_{a,N}-\ell_a}$ ,  $S_{a,N-\ell_a}$  through  $t^{\rho_a}$ , and  $\delta_a^{k-1}$  through  $t^{\widehat{v}_{a,N}}$ . Thus the valuation majorant together with (7.2) is an actual finite jet budget, not only a pole-order bound.

*Proof.* The lower bounds are

$$\text{ord}_t D_{a,e} \geq -p_{a,e}, \quad \text{ord}_t S_{e,M-\ell_e} \geq -W_{e,M-\ell_e}.$$

In a product coefficient of degree at most  $U$ , the first bound means that no coefficient of the second factor above degree  $U + p_{a,e}$  can occur; the second means that no coefficient of the first factor above degree  $U + W_{e,M-\ell_e}$  can occur. This proves (7.2). The recursion tree is finite because the remaining level decreases along every edge.

For  $B_{a,N} = R_a S_{a,N-\ell_a}$  through degree zero, the same product argument gives the stated precisions  $W_{a,N-\ell_a}$  and  $\rho_a$ . Its lower degree is at least  $-\widehat{v}_{a,N}$ , so only coefficients of  $\delta_a^{k-1}$  through degree  $\widehat{v}_{a,N}$  can enter a constant term. ■

**Lemma 7.5** (*Separated-pole Laurent cancellation*). Let  $\mu_1, \dots, \mu_s$  be distinct, and let

$$f_j(t, \Delta) \in K_\star(\Delta)((t)),$$

that is, a Laurent series in  $t$  whose coefficients are rational functions of  $\Delta$ . Suppose that every  $t$ -coefficient of  $f_j$  is a proper rational function of  $\Delta$  whose finite poles, if any, lie only at  $\Delta = \mu_j$ . If  $\sum_{j=1}^s f_j(t, \Delta)$  has no negative power of  $t$ , then neither does any  $f_j(t, \Delta)$ .

*Proof.* For each  $m < 0$ , the coefficient of  $t^m$  in the sum is zero. It is a sum of proper rational functions with pairwise disjoint pole sets. Uniqueness of partial fractions forces every summand to vanish. Repeating this for the finitely many possible negative exponents proves the claim. ■

*Proof of Theorem 4.4.* For generic  $b$ , Proposition 6.4 gives

$$H_N(c(b), \Delta, d) = \sum_{\ell_a \leq N} \frac{R_a(t)}{\Delta - h_a(t)} H_{N-\ell_a}(c(b), z_a(t), d).$$

Induction on  $M$  in the same recursion gives

$$S_{a,M}(t) = H_M(c(b), z_a(t), d)$$

for generic  $b$ . Thus

$$H_N(c(b), \Delta, d) = \sum_{\ell_a \leq N} \frac{B_{a,N}(t)}{\Delta - h_a(t)}. \quad (7.3)$$

All sums are finite.

Fix  $N$ , and choose  $\Delta$  outside  $\Lambda_N(b_*)$ . By Lemma 2.4, the inverse-Shapovalov expression is analytic in  $t$  near zero: its specialized Gram determinant is nonzero at this  $\Delta$ . Equation (7.3), initially valid for generic  $b$ , is an identity of rational functions and therefore holds on the punctured formal neighborhood of  $b_*$ . This applies for every  $\Delta$  in the complement of the finite exceptional set; hence each negative  $t$ -coefficient vanishes as a rational function of  $\Delta$ , not merely after one numerical specialization.

Group the right-hand side by its distinct limiting weights:

$$\mathcal{G}_{\lambda,N}^*(t, \Delta) = \sum_{a \in \mathcal{C}_{\lambda,N}^*} \frac{B_{a,N}(t)}{\Delta - h_a(t)}.$$

Every  $t$ -coefficient of  $\mathcal{G}_{\lambda,N}^*$  is a proper rational function of  $\Delta$  with its only possible finite pole at  $\Delta = \lambda$ . The sum over  $\lambda$  is analytic in  $t$ , as just proved. Lemma 7.5 therefore shows that each  $\mathcal{G}_{\lambda,N}^*$  is separately regular in  $t$ . Hence every collision class has a finite algebraic confluent value.

Put  $x = \Delta - \lambda$ . In the formal expansion at  $x = \infty$ ,

$$\frac{1}{\Delta - h_a(t)} = \frac{1}{x - \delta_a(t)} = \sum_{j \geq 0} \frac{\delta_a(t)^j}{x^{j+1}}.$$

Taking  $\text{CT}_t$  after summing the collision class shows that the coefficient of  $x^{-k}$  is exactly

$$\text{CT}_t \sum_{a \in \mathcal{C}_{\lambda,N}^*} B_{a,N}(t) \delta_a(t)^{k-1} = \tilde{A}_{\lambda,k,N}^*(d).$$

If  $\delta_a \not\equiv 0$ , then the summand has positive  $t$ -valuation whenever  $k > 1 + \lfloor v_{a,N}/\varepsilon_a \rfloor$ , so its constant term vanishes. If  $\delta_a \equiv 0$ , it vanishes identically for  $k > 1$ . This proves the bound  $M_{\lambda,N}^*$ , the finiteness of the pole sum, and the finite-jet nature of every constant-term calculation. Independently of cancellations, Lemma 7.2 replaces  $v_{a,N}$  by  $\widehat{v}_{a,N}$ , proving  $M_{\lambda,N}^* \leq \widehat{M}_{\lambda,N}^*$ . Lemma 7.4 then propagates the required upper precision backward through the finite recursion tree. Only finitely many terms of the formal  $x^{-1}$ -expansion can contribute to  $\text{CT}_t$ , which justifies interchanging the constant term with that expansion. The confluent value of each class tends to zero as  $\Delta \rightarrow \infty$  and has no finite pole except  $\lambda$ ; the moments are therefore its complete partial-fraction expansion, with no polynomial remainder.

Every  $R_a(t)$  is polynomial in  $d$  with coefficients in  $\mathbb{Q}(b_*)(t)$ , and all shifted denominators are independent of  $d$ . Thus every collision moment belongs to  $\mathbb{Q}(b_*)[d]$ . Equation (7.3) and the analyticity argument identify the resulting rational function with the specialized inverse-Shapovalov coefficient. Uniqueness in Theorem 4.2 gives  $\tilde{A}_{\lambda,k,N}^* = A_{\lambda,k,N}^*$ , and coefficientwise local finiteness follows from the finiteness of  $\Lambda_N(b_*)$ . ■

**Theorem 7.6** (*Pole-free Ising recursion and all-order equality*). Specialize Theorem 4.4 to  $b_\star = b_0$ , and write

$$\begin{aligned} \mathcal{C}_{n,N} &= \mathcal{C}_{\lambda_n,N}^\star, & M_{n,N} &= M_{\lambda_n,N}^\star, & \widehat{M}_{n,N} &= \widehat{M}_{\lambda_n,N}^\star, & m_{n,N} &= m_{\lambda_n,N}^\star, \\ \widetilde{A}_{n,k,N} &= \widetilde{A}_{\lambda_n,k,N}^\star. \end{aligned}$$

For every  $N \geq 1$ ,

$$H_N(1/2, \Delta, d) = \sum_{\substack{n \geq 0 \\ \mathcal{C}_{n,N} \neq \emptyset}} \sum_{k=1}^{M_{n,N}} \frac{\widetilde{A}_{n,k,N}(d)}{(\Delta - \lambda_n)^k}. \quad (7.4)$$

The sums and every constant-term calculation are finite,  $\widetilde{A}_{n,k,N}(d) \in \mathbb{Q}[d]$ , and the recursion in Definitions 7.1–7.3 terminates after levels strictly below  $N$ . In particular, by the uniqueness in Theorem 4.3, the collision moments are the partial-fraction coefficients of the block,

$$\widetilde{A}_{n,k,N} = A_{n,k,N},$$

and  $m_{n,N}$  is the exact multiplicity of the pole of  $H_N$  at  $\lambda_n$ . Moreover

$$m_{n,N} \leq M_{n,N} \leq \widehat{M}_{n,N},$$

where  $\widehat{M}_{n,N}$  is computed before simplifying cancellations inside the shifted blocks. The backward rule of Lemma 7.4 supplies a finite truncation order for every intermediate Laurent germ. Consequently, with  $A_{n,k}(d; q)$  as in Theorem 4.3, one has the coefficientwise locally finite identity

$$H_\Delta^{(1/2)}(d \mid q) = 1 + \sum_{n \geq 0} \sum_{k \geq 1} \frac{A_{n,k}(d; q)}{(\Delta - \lambda_n)^k}.$$

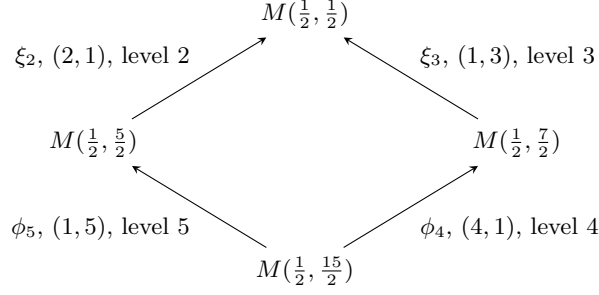
This series agrees coefficient by coefficient with Definition 2.2, hence with the inverse-Shapovalov definition of the torus one-point block.

*Proof.* Proposition 3.3 identifies the general collision classes with the displayed  $\mathcal{C}_{n,N}$ . The specialized inverse-Shapovalov coefficient and all pole positions  $\lambda_n$  have rational coefficients, so uniqueness of partial fractions improves  $\widetilde{A}_{n,k,N} \in \mathbb{Q}(b_0)[d]$  to  $\widetilde{A}_{n,k,N} \in \mathbb{Q}[d]$ . All other assertions are the specialized assertions of Theorem 4.4.  $\blacksquare$

## 8. The energy-module level-seven diamond

This section and the next work out the two embedding diamonds of Figures 1 and 2 and prove the corresponding fixed- $c$  principal-part formulas; Table 1 summarizes the resulting local data.

**Proposition 8.1** (*First nontransverse Smith layer*). Let  $h_E = 1/2$ , and let  $U_2, U_3$  be the submodules generated by  $\xi_2, \xi_3$  in Proposition 5.7. Their first intersection is generated



**Figure 1.** The energy diamond. Arrows are the singular-vector embeddings of Proposition 8.1, labelled by the embedded singular vector, its Kac label, and its relative level. The two branches meet first at total level 7, where  $\xi_7 = j_{\xi_2}(\phi_5) = j_{\xi_3}(\phi_4)$  creates the first double pole of the Ising block.

at total level 7 by a normalized singular vector  $\xi_7$ . More explicitly, the normalized child singular vectors are

$$\phi_5 = (L_{-1}^5 - 15L_{-2}L_{-1}^3 + 36L_{-2}^2L_{-1} + \frac{99}{4}L_{-3}L_{-1}^2 - 45L_{-3}L_{-2} - 54L_{-4}L_{-1} + 45L_{-5})|\frac{5}{2}\rangle,$$

$$\phi_4 = (L_{-1}^4 - \frac{40}{3}L_{-2}L_{-1}^2 + 16L_{-2}^2 + \frac{88}{3}L_{-3}L_{-1} - \frac{152}{3}L_{-4})|\frac{7}{2}\rangle,$$

and

$$\xi_7 = j_{\xi_2}(\phi_5) = j_{\xi_3}(\phi_4).$$

The 15 local Smith exponents of  $G_7(1/2, h_E + x)$  are

$$(0^4, 1^{10}, 2).$$

For the second crossing form of Proposition 5.3, normalized by the displayed  $\xi_7$ ,

$$\gamma_7 = -700700.$$

*Proof.* The two child highest weights are

$$h_E + 2 = \frac{5}{2} = \lambda_{11}, \quad h_E + 3 = \frac{7}{2} = \lambda_{13}.$$

Their first singular vectors have relative levels 5 and 4, with labels (1, 5) and (4, 1), respectively. Direct Virasoro commutation checks the two displayed vectors. Their images are nonzero singular vectors at total level 7; uniqueness and their common  $L_{-1}^7$ -coefficient 1 prove the displayed equality. The embedding diagram in [IK11, Proposition 5.4 and Figure 5.4] also shows that there is no earlier intersection and that the radical is the sum of the two singular submodules.

At level 7, the radical therefore has dimension

$$p(5) + p(4) - 1 = 7 + 5 - 1 = 11.$$



The only Kac labels of weight  $1/2$  and level at most 7 are  $(2, 1)$  and  $(1, 3)$ , so the determinant valuation is

$$p(5) + p(4) = 12.$$

The Smith exponents are consequently  $(0^4, 1^{10}, 2)$ .

For completeness, the second crossing scalar can be obtained without an off-critical deformation. In the level-seven PBW basis, form

$$G_7(1/2, h_E + x) = G_0 + xG_1 + x^2G_2 + O(x^3),$$

solve  $G_0v_1 = -G_1\xi_7$ , and evaluate

$$\xi_7^\top(G_2\xi_7 + G_1v_1).$$

Exact PBW commutation gives  $-700700$ . Choice-independence follows from Proposition 5.3. ■

**Proposition 8.2** (*No earlier nontransverse pole*). *For every  $N < 7$  and every finite internal weight, all local Smith exponents of  $G_N(1/2, \Delta)$  are at most 1. Consequently  $G_N^{-1}$ ,  $F_N$ , and  $H_N$  have at most simple poles in  $\Delta$ . Together with Proposition 8.1, this proves that the double pole at  $\Delta = 1/2$  and level 7 is the first double pole of the Ising Verma block.*

*Proof.* For a Kac weight represented by only one label  $a = (r, s)$  with  $\ell_a \leq N$ , injectivity of the singular-vector embedding gives a radical of dimension  $p(N - \ell_a)$ . This equals the valuation contributed by that label to the Kac determinant. Lemma 5.1 therefore makes all corresponding positive Smith exponents equal to 1.

It remains to inspect repeated labels. Proposition 3.3, or direct enumeration of  $rs \leq 6$ , shows that the complete list of repeated classes below level 7 is

| weight                     | labels           | singular levels |
|----------------------------|------------------|-----------------|
| $\lambda_5 = \frac{1}{2}$  | $(2, 1), (1, 3)$ | 2, 3            |
| $\lambda_2 = \frac{1}{16}$ | $(1, 2), (2, 2)$ | 2, 4            |
| $\lambda_1 = 0$            | $(1, 1), (2, 3)$ | 1, 6.           |

The two energy branches have no intersection below level 7 by Proposition 8.1.

For the spin class, the child highest weights along the level-two and level-four branches are  $33/16$  and  $65/16$ . Their Kac determinants have no zero below relative levels 8 and 6, respectively. A first intersection below level 7 would be singular; injectivity would pull it back along the level-two branch to a singular vector of relative level at most 4, which is impossible.

For the vacuum class, the level-one branch is generated by  $L_{-1}|0\rangle$  and has child highest weight 1. The Kac determinant of  $M(1/2, 1)$  has no zero through relative level 5. Hence the level-six singular vector cannot lie in the level-one branch, since its preimage would be singular at relative level 5.

Thus, at every repeated weight and every  $N < 7$ , the relevant singular submodules have zero intersection. The radical dimension is therefore the sum  $\sum_a p(N - \ell_a)$ , exactly the Kac-determinant valuation. A final application of Lemma 5.1 proves the Smith and pole bounds. ■

**Theorem 8.3** (*First nontransverse fixed-c principal part*). Define

$$E_5(d) = \mathcal{P}_{1,5}(b_0, d), \quad E_4(d) = \mathcal{P}_{4,1}(b_0, d).$$

Explicitly,

$$E_5(d) = \frac{1}{16}d(d-20)(d-13)(d-11)(d-6)(d-1)(2d-15)(2d-7)(2d-5)(2d-1),$$

$$E_4(d) = \frac{1}{9}d(d-20)(d-13)(d-11)(d-6)(d-1)(3d-14)(3d-5),$$

and  $E_2E_5 = E_3E_4$ . Put

$$Q_5^E(d) = \left. \frac{\partial}{\partial z} \left\{ \left( z - \frac{5}{2} \right) H_5(1/2, z, d) \right\} \right|_{z=5/2},$$

$$Q_4^E(d) = \left. \frac{\partial}{\partial z} \left\{ \left( z - \frac{7}{2} \right) H_4(1/2, z, d) \right\} \right|_{z=7/2}.$$

For

$$\alpha = (2, 1), \quad \beta = (1, 3), \quad \mu = (1, 5), \quad \nu = (4, 1),$$

let

$$\delta_\alpha = h_\alpha + 2 - h_\mu, \quad \delta_\beta = h_\beta + 3 - h_\nu,$$

and, with dots denoting  $d/db$  at  $b_0$ , define the explicit finite polynomial

$$\Theta_7(d) = \sum_{(\kappa, \tau) = (\alpha, \mu), (\beta, \nu)} \left( \frac{\dot{\mathcal{R}}_\kappa \mathcal{R}_\tau + \mathcal{R}_\kappa \dot{\mathcal{R}}_\tau}{\dot{\delta}_\kappa} - \frac{\mathcal{R}_\kappa \mathcal{R}_\tau \ddot{\delta}_\kappa}{2\dot{\delta}_\kappa^2} \right)_{b=b_0}.$$

Its factored expansion is recorded in equation (A.1) of Appendix A. Then

$$\begin{aligned} \text{PP}_{\Delta=1/2} H_7(1/2, \Delta, d) = & - \frac{E_2(d)E_5(d)}{700700(\Delta - \frac{1}{2})^2} \\ & + \frac{\frac{9}{28}E_2(d)Q_5^E(d) - \frac{8}{105}E_3(d)Q_4^E(d) + \Theta_7(d)}{\Delta - \frac{1}{2}}. \end{aligned} \quad (8.1)$$

This is the earliest double pole of the Ising block.

*Proof.* By Theorem 7.6, the confluent Laurent expansion computed below coincides with the fixed-c Laurent expansion of  $H_7$  at  $\Delta = \frac{1}{2}$ . The double-pole coefficient also has an intrinsic derivation that avoids the confluence altogether: Propositions 5.3 and 8.1 give  $[x^{-2}]G_7(1/2, \frac{1}{2} + x)^{-1} = \xi_7 \xi_7^\top / \gamma_7$ , Lemma 5.5 applied along either branch of the diamond gives  $\langle \xi_7 | V_d(1) | \xi_7 \rangle = E_2E_5 = E_3E_4$ , and multiplication by the Euler product does not affect the  $x^{-2}$  coefficient at level seven because all lower levels have at most simple poles by Proposition 8.2.

Equation (6.2) gives the four fusion polynomials. Direct evaluation of the residue products and Kac slopes gives

|                          |          |          |             |           |
|--------------------------|----------|----------|-------------|-----------|
| $(r, s)$                 | $(2, 1)$ | $(1, 3)$ | $(1, 5)$    | $(4, 1)$  |
| $\mathcal{A}_{r,s}(b_0)$ | $9/28$   | $-8/105$ | $-1/102375$ | $3/61600$ |

and

$$\frac{\dot{h}_{2,1} - \dot{h}_{1,3}}{\dot{h}_{2,1} - \dot{h}_{1,5}} = \frac{5}{11}, \quad \frac{\dot{h}_{1,3} - \dot{h}_{2,1}}{\dot{h}_{1,3} - \dot{h}_{4,1}} = \frac{5}{13}.$$

Either path therefore expresses the coefficient produced jointly by their cancelled  $t^{-1}$  parts:

$$\frac{5}{11} \mathcal{A}_{2,1} \mathcal{A}_{1,5} = \frac{5}{13} \mathcal{A}_{1,3} \mathcal{A}_{4,1} = -\frac{1}{700700}.$$

This agrees with the intrinsic inverse second crossing scalar  $\gamma_7^{-1}$  from Proposition 8.1.

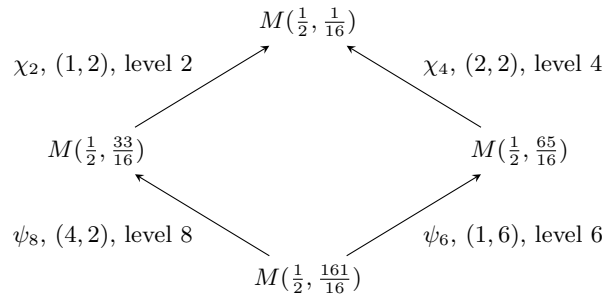
Expanding the two outer recursion terms at  $\Delta = 1/2$ , their cancelled  $t^{-1}$  numerators give the displayed double pole. Their finite parts are respectively

$$\frac{9}{28} E_2 Q_5^E + T_{\alpha,\mu}, \quad -\frac{8}{105} E_3 Q_4^E + T_{\beta,\nu},$$

where the two  $T$ 's are precisely the summands defining  $\Theta_7$ . Differentiating the finite products of Definition 6.1 and factoring over  $\mathbb{Q}[d]$  gives the displayed explicit polynomial form of  $\Theta_7$ . This proves the complete principal part. Its global minimality in level is Proposition 8.2.  $\blacksquare$

*Remark 8.4* (Computing the finite parts). The lower-level finite parts  $Q_5^E, Q_4^E$  here, and  $Q_8, Q_6$  in Theorem 9.4 below, are defined by fixed- $c$  data only: they can be computed directly from the inverse-Shapovalov matrices or recursively from Theorem 7.6, and no off-critical numerical limit is involved. Closed forms (of degrees 16 and 12 in  $d$  for  $Q_8$  and  $Q_6$ , with unwieldy rational coefficients) are recorded in machine-readable form and verified against the inverse-Shapovalov definition by the audit scripts; see Remark 9.6.

## 9. The spin-module level-ten diamond



**Figure 2.** The spin diamond. The two branches meet first at total level 10, where  $\chi_{10} = j_{\chi_2}(\psi_8) = j_{\chi_4}(\psi_6)$  creates the double pole of Theorem 9.4.

**Proposition 9.1** (*Level-ten Smith profile at  $\Delta = 1/16$* ). Let  $h_0 = 1/16$ , and let

$$U_2 = U(\text{Vir}_-)\chi_2, \quad U_4 = U(\text{Vir}_-)\chi_4$$

be the two singular submodules of  $M(1/2, h_0)$  from Proposition 5.10. Their first intersection is generated at total level 10 by a singular vector  $\chi_{10}$ , and

$$\dim \ker G_{10}(1/2, h_0) = 32.$$

The 42 local Smith exponents of  $G_{10}(1/2, h_0 + x)$  are

$$(0^{10}, 1^{31}, 2).$$

In particular, the inverse Shapovalov matrix has a nonzero rank-one coefficient at order  $x^{-2}$ .

*Proof.* The weight  $h_0$  is the interior Ising weight labelled by  $(1, 2)$ , so it is in Class  $R^+$ , Case 1+, in the classification of [IK11, Section 5.1.4]: both Coulomb-gas parameters are rational and the weight lies in the interior of the Kac table, so the embedding diagram is the standard braided pattern of two infinite strings of singular vectors. The commutative embedding diagram in [IK11, Proposition 5.4 and Figure 5.4] describes the two branches and their successive intersections. The identification of the first Jantzen layer with their sum, hence with the maximal proper submodule, is [IK11, Theorem 6.3(i) and equation (6.3)].

The two highest weights are

$$h_0 + 2 = \frac{33}{16} = \lambda_{10}, \quad h_0 + 4 = \frac{65}{16} = \lambda_{14}.$$

The first singular vector inside  $U_2$  has relative level 8, with Kac label  $(4, 2)$ , and the first singular vector inside  $U_4$  has relative level 6, with Kac label  $(1, 6)$ . Both therefore have total level 10 and weight

$$h_0 + 10 = \frac{161}{16} = \lambda_{22}.$$

The embeddings are injective by [IK11, the paragraph preceding Corollary 5.2]. Their two images are proportional by uniqueness of a singular vector at a fixed level, [IK11, Proposition 5.1]. Hence the level-ten intersection is one dimensional. There is no earlier intersection: a vector at the first intersection level would be singular, whereas the Kac determinant for  $M(1/2, 33/16)$ , respectively  $M(1/2, 65/16)$ , has no zero below relative level 8, respectively 6.

The level-ten radical is the level-ten part of  $U_2 + U_4$ . Since

$$p(8) = 22, \quad p(6) = 11,$$

and the intersection has dimension one at this level,

$$\dim \ker G_{10}(1/2, h_0) = 22 + 11 - 1 = 32.$$

By Proposition 3.3, the only positive Kac labels with  $\Delta_{r,s} = h_0$  and  $rs \leq 10$  are  $(1, 2)$  and  $(2, 2)$ . The Kac determinant formula therefore gives

$$\text{ord}_x \det G_{10}(1/2, h_0 + x) = p(8) + p(6) = 33.$$

There are  $p(10) = 42$  Smith exponents, exactly 32 of them are positive, and their sum is 33. Thus 31 positive exponents equal 1, one equals 2, and the remaining 10 vanish. Lemma 5.1 gives the final assertion. (This illustrates Corollary 5.4: the determinant valuation 33 records the sum of the positive Smith exponents, while the pole order is their maximum, 2.) ■

**Lemma 9.2** (*Exact constants of the level-ten confluence*). Let  $b_0 = 2i/\sqrt{3}$ , so  $c(b_0) = 1/2$ . Then

$$\frac{(r, s)}{\mathcal{A}_{r,s}(b_0)} \mid \begin{array}{cccc} (1, 2) & (2, 2) & (4, 2) & (1, 6) \\ -4/7 & 648/13475 & 48/32035128125 & -8/320945625. \end{array}$$

If a dot denotes  $d/db$  at  $b_0$ , then

$$\frac{\dot{h}_{1,2} - \dot{h}_{2,2}}{\dot{h}_{1,2} - \dot{h}_{4,2}} = \frac{1}{5}, \quad \frac{\dot{h}_{2,2} - \dot{h}_{1,2}}{\dot{h}_{2,2} - \dot{h}_{1,6}} = \frac{1}{7},$$

and

$$\frac{1}{5}\mathcal{A}_{1,2}\mathcal{A}_{4,2} = \frac{1}{7}\mathcal{A}_{2,2}\mathcal{A}_{1,6} = -\frac{192}{1121229484375}.$$

Moreover,

$$\begin{aligned} \mathcal{P}_{4,2}(b_0, d) &= \frac{1}{20736} d(d-20)(d-13)(d-11)(d-6)(d-1) \\ &\quad \times (2d-57)(2d-35)(2d-15)(2d-7)(2d-5)(2d-1) \\ &\quad \times (3d-14)(3d-5)(6d-55)(6d-1), \end{aligned}$$

$$\begin{aligned} \mathcal{P}_{1,6}(b_0, d) &= \frac{1}{64} d(d-20)(d-13)(d-11)(d-6)(d-1) \\ &\quad \times (2d-57)(2d-35)(2d-15)(2d-7)(2d-5)(2d-1), \end{aligned}$$

and

$$P_2(d)\mathcal{P}_{4,2}(b_0, d) = P_4(d)\mathcal{P}_{1,6}(b_0, d).$$

*Proof.* Substitute  $b_0 = 2i/\sqrt{3}$  in the finite products of Definition 6.1. Direct differentiation gives

$$c(b_0) = \frac{1}{2}, \quad c'(b_0) = \frac{7i\sqrt{3}}{2} \neq 0,$$

so  $b - b_0$  is a valid local deformation parameter. The Kac-weight derivatives are

$$\frac{(r, s)}{16h_{r,s}/(i\sqrt{3})} \mid \begin{array}{cccc} (1, 2) & (2, 2) & (4, 2) & (1, 6) \\ 9 & -7 & -71 & 105. \end{array}$$

These four values give the two slope ratios. Multiplication of the four displayed  $\mathcal{A}$ -values gives the common nested residue. Factoring the two  $\mathcal{P}$ -products over  $\mathbb{Q}[d]$  gives the last three identities. ■

**Theorem 9.3** (*Leading double-pole coefficient at level ten*). Normalize all singular vectors by setting their  $L_{-1}^\ell$  coefficient equal to 1. Let  $\psi_8 \in M(1/2, 33/16)$  and  $\psi_6 \in M(1/2, 65/16)$  be the singular vectors of levels 8 and 6, respectively. Then

$$\chi_{10} = j_{\chi_2}(\psi_8) = j_{\chi_4}(\psi_6).$$

Define

$$P_8^{(33/16)}(d) = \langle \psi_8 | V_d(1) | \psi_8 \rangle, \quad P_6^{(65/16)}(d) = \langle \psi_6 | V_d(1) | \psi_6 \rangle.$$

They are the explicit polynomials

$$P_8^{(33/16)}(d) = \mathcal{P}_{4,2}(b_0, d), \quad P_6^{(65/16)}(d) = \mathcal{P}_{1,6}(b_0, d)$$

listed in Lemma 9.2. Write

$$G_{10}(1/2, h_0 + x) = G_0 + xG_1 + x^2G_2 + O(x^3),$$

choose  $v_1$  with

$$G_0 v_1 = -G_1 \chi_{10},$$

and set

$$\gamma_{10} = \chi_{10}^T (G_2 \chi_{10} + G_1 v_1).$$

Then it is independent of the choice of  $v_1$ , and

$$\gamma_{10} = -\frac{1121229484375}{192}.$$

Consequently,

$$\lim_{\Delta \rightarrow h_0} (\Delta - h_0)^2 H_{10}(1/2, \Delta, d) = \frac{P_2(d) P_8^{(33/16)}(d)}{\gamma_{10}} = \frac{P_4(d) P_6^{(65/16)}(d)}{\gamma_{10}}.$$

This is an intrinsic fixed- $c$  expression for the spin-module nontransverse Laurent coefficient.

*Proof.* The two compositions in the first display are nonzero singular vectors of total level 10. Proposition 5.1 of [IK11] makes them proportional. Their  $L_{-1}^{10}$  coefficients are both 1, so they are equal. The Kac determinants of the child modules of highest weights 33/16 and 65/16 have no zero below levels 8 and 6, respectively. Consequently the standard  $L_{-1}^\ell$ -normalized singular-vector recursion has a nonsingular coefficient system at every preceding level. Cramer's rule then shows that the normalized rational families  $\psi_8(b)$  and  $\psi_6(b)$  are regular at  $b_0$ . Hence the rational identity (6.2) specializes to

$$P_8^{(33/16)} = \mathcal{P}_{4,2}(b_0, d), \quad P_6^{(65/16)} = \mathcal{P}_{1,6}(b_0, d).$$

Let  $K_1 = \ker G_0 = (U_2 + U_4)_{10}$ , and let  $\beta_1$  be its first crossing form. By Lemma 5.6, the restriction of  $\beta_1$  to  $U_2$  is the scalar  $b_{\chi_2}$  times the Shapovalov form on  $M(1/2, 33/16)$ . Since  $\psi_8$  is singular,  $\beta_1(\chi_{10}, U_2) = 0$ . The same argument through  $U_4$  gives  $\beta_1(\chi_{10}, U_4) = 0$ . Therefore  $\chi_{10} \in K_2 = \ker \beta_1$ .

Proposition 9.1 shows that exactly one Smith exponent is at least 2. The unconditional Smith-layer count in Proposition 5.3 therefore shows that  $K_2$  is spanned by  $\chi_{10}$ . Since that exponent is exactly 2, the same proposition shows that its second crossing form is the nonzero scalar  $\gamma_{10}$ , and gives

$$[x^{-2}]G_{10}(1/2, h_0 + x)^{-1} = \frac{\chi_{10} \chi_{10}^T}{\gamma_{10}}.$$

Taking the trace against the specialized torus matrix yields

$$[x^{-2}]F_{10}(1/2, h_0 + x, d) = \frac{\langle \chi_{10} | V_d(1) | \chi_{10} \rangle}{\gamma_{10}}.$$

For  $N < 10$ , the two singular submodules have zero intersection at level  $N$ . Thus the radical dimension is

$$p(N-2) + p(N-4),$$

with  $p(k) = 0$  for  $k < 0$ , and this equals the Kac-determinant valuation. Lemma 5.1 therefore shows that all lower-level Smith exponents at  $h_0$  are at most one. Multiplication by the Euler product consequently does not alter the  $x^{-2}$  coefficient at level 10. Finally, Lemma 5.5, applied along either side of the embedding diamond, gives

$$\langle \chi_{10} | V_d(1) | \chi_{10} \rangle = P_2(d) P_8^{(33/16)}(d) = P_4(d) P_6^{(65/16)}(d).$$

The Gram matrices and the normalized singular-vector coefficients are rational at  $c = 1/2$ , so the defining linear system gives  $\gamma_{10} \in \mathbb{Q}$ . It remains to evaluate this intrinsic scalar. Apply the generic recursion of Definition 6.1 algebraically near  $b_0$ . By Theorem 7.6, once the resonant paths are combined, their constant term equals the specialized fixed- $c$  coefficient as a rational function of  $\Delta$ , including the coefficient of  $(\Delta - h_0)^{-2}$ .

The two nested paths have opposite  $t^{-1}$  numerator coefficients. After their cancellation, the resulting double-pole coefficient can be written using the  $(1, 2) \rightarrow (4, 2)$  path as

$$\frac{\dot{h}_{1,2} - \dot{h}_{2,2}}{\dot{h}_{1,2} - \dot{h}_{4,2}} \mathcal{A}_{1,2}(b_0) \mathcal{A}_{4,2}(b_0) P_2(d) P_8^{(33/16)}(d).$$

Lemma 9.2 evaluates this scalar coefficient as  $-192/1121229484375$ . Rewriting the same combined coefficient using the other path gives the equal expression with slope ratio  $1/7$ ; the two expressions are alternatives, not additive contributions. Comparison with the already proved intrinsic coefficient  $P_2 P_8 / \gamma_{10}$  yields the displayed value of  $\gamma_{10}$ . ■

**Theorem 9.4** (*Finite recursive level-ten principal part*). Define the finite shifted-block parts

$$Q_8(d) = \frac{\partial}{\partial z} \left\{ \left( z - \frac{33}{16} \right) H_8(1/2, z, d) \right\} \Big|_{z=33/16}$$

and

$$Q_6(d) = \frac{\partial}{\partial z} \left\{ \left( z - \frac{65}{16} \right) H_6(1/2, z, d) \right\} \Big|_{z=65/16}.$$

For

$$\alpha = (1, 2), \quad \beta = (2, 2), \quad \mu = (4, 2), \quad \nu = (1, 6),$$

set

$$\delta_\alpha = h_\alpha + 2 - h_\mu, \quad \text{quad} \delta_\beta = h_\beta + 4 - h_\nu,$$

and, with dots denoting  $d/db$  at  $b_0$ , put

$$\Theta_{10}(d) = \sum_{(\kappa, \tau) = (\alpha, \mu), (\beta, \nu)} \left( \frac{\dot{\mathcal{R}}_\kappa \mathcal{R}_\tau + \mathcal{R}_\kappa \dot{\mathcal{R}}_\tau}{\dot{\delta}_\kappa} - \frac{\mathcal{R}_\kappa \mathcal{R}_\tau \ddot{\delta}_\kappa}{2\dot{\delta}_\kappa^2} \right)_{b=b_0}.$$

Its factored expansion is equation (A.2) of Appendix A. Then the entire principal part of the level-ten reduced coefficient at  $\Delta = h_0 = 1/16$  is

$$\begin{aligned} \text{PP}_{\Delta=h_0} H_{10}(1/2, \Delta, d) = & -\frac{192 P_2(d) P_8^{(33/16)}(d)}{1121229484375 (\Delta - h_0)^2} \\ & + \frac{-\frac{4}{7} P_2(d) Q_8(d) + \frac{648}{13475} P_4(d) Q_6(d) + \Theta_{10}(d)}{\Delta - h_0}. \end{aligned} \quad (9.1)$$

Every term on the right is finite and is evaluated at  $c = 1/2$ . Thus this is a complete finite recursive formula for the principal part.

*Proof.* Let  $t = b - b_0$ , and abbreviate the four labels

$$\alpha = (1, 2), \quad \beta = (2, 2), \quad \mu = (4, 2), \quad \nu = (1, 6).$$

Set

$$z_\alpha = h_\alpha + 2, \quad z_\beta = h_\beta + 4, \quad \delta_\alpha = z_\alpha - h_\mu, \quad \delta_\beta = z_\beta - h_\nu.$$

Only the  $\alpha$ - and  $\beta$ -terms of the generic recursion can develop a pole at  $\Delta = h_0$ . Their numerators at level ten are

$$C_\alpha(b) = \mathcal{R}_\alpha(b, d) H_8(c(b), z_\alpha(b), d), \quad C_\beta(b) = \mathcal{R}_\beta(b, d) H_6(c(b), z_\beta(b), d).$$

At  $(b_0, z_\alpha(b_0))$ , the level-eight shifted block has one transverse pole, labelled by  $\mu$ , and no other Kac divisor through that point. The analogous statement holds for the level-six block and  $\nu$ . Therefore the definitions of  $Q_8, Q_6$  give

$$\begin{aligned} H_8(c(b), z_\alpha(b), d) &= \frac{\mathcal{R}_\mu(b, d)}{\delta_\alpha(b)} + Q_8(d) + O(t), \\ H_6(c(b), z_\beta(b), d) &= \frac{\mathcal{R}_\nu(b, d)}{\delta_\beta(b)} + Q_6(d) + O(t). \end{aligned}$$

For either nested pair  $(\kappa, \tau)$ , expansion of  $\mathcal{R}_\kappa \mathcal{R}_\tau / \delta_\kappa$  gives finite term

$$T_{\kappa, \tau} = \frac{\dot{\mathcal{R}}_\kappa \mathcal{R}_\tau + \mathcal{R}_\kappa \dot{\mathcal{R}}_\tau}{\dot{\delta}_\kappa} - \frac{\mathcal{R}_\kappa \mathcal{R}_\tau \ddot{\delta}_\kappa}{2 \dot{\delta}_\kappa^2},$$

where every undotted factor on the right is evaluated at  $b_0$ . The two  $t^{-1}$  terms cancel because

$$\frac{\mathcal{R}_\alpha \mathcal{R}_\mu}{\dot{\delta}_\alpha} + \frac{\mathcal{R}_\beta \mathcal{R}_\nu}{\dot{\delta}_\beta} = 0;$$

this is Lemma 9.2 together with  $P_2 P_8 = P_4 P_6$ .

Write  $y = \Delta - h_0$ . Expanding

$$\frac{C_\alpha(b)}{\Delta - h_\alpha(b)} + \frac{C_\beta(b)}{\Delta - h_\beta(b)}$$

at  $t = 0$ , the cancelled  $t^{-1}$  numerator produces the  $y^{-2}$  coefficient

$$\frac{\dot{h}_\alpha - \dot{h}_\beta}{\dot{\delta}_\alpha} \mathcal{R}_\alpha(b_0, d) \mathcal{R}_\mu(b_0, d) = -\frac{192}{1121229484375} P_2(d) P_8^{(33/16)}(d).$$



The  $y^{-1}$  coefficient is

$$\mathcal{R}_\alpha(b_0, d)Q_8(d) + \mathcal{R}_\beta(b_0, d)Q_6(d) + T_{\alpha, \mu} + T_{\beta, \nu}.$$

The first two residue constants are  $-4P_2/7$  and  $648P_4/13475$ . Differentiating the finite products in Definition 6.1 and factoring over  $\mathbb{Q}[d]$  gives

$$T_{\alpha, \mu} + T_{\beta, \nu} = \Theta_{10}(d).$$

Substitution proves the stated principal part. ■

**Corollary 9.5** (*Recovery of the explicit confluent formulas*). *The  $N = 3$  and  $N = 7$  moments in the  $\lambda_5 = 1/2$  class give Theorems 5.8 and 8.3. The  $N = 4$  and  $N = 10$  moments in the  $\lambda_2 = 1/16$  class give Theorems 5.11 and 9.4.*

*Proof.* At  $N = 3$ , the class  $\mathcal{C}_{5,3}$  consists of  $(2, 1)$  and  $(1, 3)$ . Its first moment is

$$\frac{9}{28}E_2(d)H_1(1/2, 5/2, d) - \frac{8}{105}E_3(d),$$

and its higher moments vanish. At  $N = 7$ , the same class has the two nested shifted poles  $(2, 1) \rightarrow (1, 5)$  and  $(1, 3) \rightarrow (4, 1)$ . Its first two moments are the simple- and double-pole coefficients of Theorem 8.3.

At  $N = 4$ , the class  $\mathcal{C}_{2,4}$  consists of  $(1, 2)$  and  $(2, 2)$ , and both  $B$ -terms are regular at  $t = 0$ . Its first moment is

$$-\frac{4}{7}P_2(d)H_2(1/2, 33/16, d) + \frac{648}{13475}P_4(d),$$

while all higher moments vanish.

At  $N = 10$ , the same spin collision class has

$$B_{(1,2),10} = \mathcal{R}_{1,2}H_8(c(b), h_{1,2} + 2, d), \quad B_{(2,2),10} = \mathcal{R}_{2,2}H_6(c(b), h_{2,2} + 4, d).$$

Their zeroth and first collision moments are exactly the two Laurent coefficients computed in the proof of Theorem 9.4; all higher moments vanish by the level-ten Smith profile. ■

*Remark 9.6* (Exact replay and its scope). The PBW commutation, Ward recursion, determinant, rank, and null-vector checks for the spin collision in Propositions 5.9 and 5.10, as well as every residue identity in Theorem 5.11, are replayed by `tools/audit_low_levels.py`. The computation uses exact rational arithmetic. The singular vectors, fusion polynomials,  $15 \times 15$  Gram matrix, Smith-layer dimensions, and intrinsic second crossing scalar for the earlier energy-module level-three and level-seven resonances are replayed by `tools/audit_first_resonances.py`. The same script enumerates every repeated Kac class below level seven and verifies the child-module determinant gaps used in Proposition 8.2. The Kac slopes, four generic residue constants, fusion factorization, generic-recursion evaluation of  $\gamma_{10}$ , and polynomial  $\Theta_{10}$  are replayed by `tools/audit_grade10_confluence.py`. The shifted Laurent recursion and its first two collision moments are independently replayed by

|                                | energy class              | spin class                                 |
|--------------------------------|---------------------------|--------------------------------------------|
| collided weight                | $\lambda_5 = \frac{1}{2}$ | $\lambda_2 = \frac{1}{16}$                 |
| colliding labels               | $(2, 1), (1, 3)$          | $(1, 2), (2, 2)$                           |
| parent singular levels         | 2, 3                      | 2, 4                                       |
| first collided coefficient     | $H_3$                     | $H_4$                                      |
| first residue                  | Theorem 5.8               | Theorem 5.11                               |
| diamond closes at level        | 7                         | 10                                         |
| child labels (relative levels) | $(1, 5) (5), (4, 1) (4)$  | $(4, 2) (8), (1, 6) (6)$                   |
| second crossing scalar         | $\gamma_7 = -700700$      | $\gamma_{10} = -\frac{1121229484375}{192}$ |
| principal part                 | Theorem 8.3               | Theorem 9.4                                |

**Table 1.** The two Ising embedding diamonds and their local data.

`tools/audit_confluent_recursion.py`. The level-ten confluence script checks the algebraic confluence calculation; it does not construct the  $42 \times 42$  Gram matrix,  $\chi_{10}$ , or the second crossing form independently. The Smith profile at level ten follows from the determinant and embedding-diagram argument in Proposition 9.1.

Two further scripts test the principal-part theorems directly against the inverse-Shapovalov definition rather than against the confluence algebra. `tools/audit_direct_principal_parts.py` reconstructs  $H_N$ ,  $N \leq 7$ , exactly as rational functions of  $\Delta$  at sample external weights and verifies Theorems 5.8 and 5.11, the complete level-seven principal part of Theorem 8.3 including the explicit  $\Theta_7$ , the pole-order claims of Proposition 8.2, the table of Section 10, and the closed forms of  $Q_4^E, Q_5^E$ . `tools/audit_grade10_direct.py` performs the analogous direct  $42 \times 42$  computation at level ten: it verifies the kernel dimension and determinant valuation of Proposition 9.1 and the complete principal part of Theorem 9.4, including the value of  $\gamma_{10}$  and the closed forms of  $Q_8, Q_6$ . Finally, `tools/audit_stocco_comparison.py` verifies the imported identities of Remark 6.2 at a generic sample point and the agreement with the pole-free expressions of [Sto22] discussed in Section 10.

## 10. Low-level output and comparison with earlier expansions

Through level four, every pole of the Ising block is simple:  $m_{n,N} = 1$  for all classes with  $N \leq 4$ , by Proposition 8.2. The complete list of nonzero collision moments through level four, computed from Definition 7.3 and equal to the partial-fraction coefficients  $A_{n,k,N}$  by Theorem 7.6, is factored over  $\mathbb{Q}[d]$  in equation (A.3) of Appendix A. The entries are consistent with the local theorems:  $A_{5,1,3}$  is the residue of Theorem 5.8,  $A_{2,1,4}$  is the polynomial of Theorem 5.11, and  $A_{13,1,4} = \mathcal{A}_{4,1}(b_0)\mathcal{P}_{4,1}(b_0, d)$ . All entries are verified against the inverse-Shapovalov definition by `tools/audit_direct_principal_parts.py`.

*Remark 10.1* (Comparison with the pole-free expressions of Stocco). Stocco [Sto22, Sections II.2–II.3 and Appendix A] writes the generic torus recursion over a common denominator,

$$H_{\Delta}^{(N)} = \frac{R_{1,1} K_N(\Delta, \Delta_1, \beta)}{\prod_{kl \leq N} (\Delta - \Delta_{\langle k, l \rangle})},$$

and gives closed pole-free numerators  $K_N$  for  $N \leq 4$ : order two in equation (II.11), orders three and four in Appendix A. In our notation  $\beta = b^2$ ,  $\Delta_1 = d$ , and his  $\Delta_{\langle r,s \rangle}$  is  $h_{r,s}(b)$ . We have verified in exact arithmetic that his order-two and order-three numerators reproduce the generic blocks  $H_2$  and  $H_3$  of Definition 2.2 identically in  $b$ , and that his order-four numerator reproduces  $H_4$  at sample values of  $b$  and at  $b = b_0$ ; see `tools/audit_stocco_comparison.py`. (The identical-in- $b$  comparison was run only through order three because of the cost of expanding the order-four rational functions; the sampled order-four check includes the specialization  $b = b_0$  used in this paper.) In particular, specializing his expressions at  $\beta = b_0^2 = -4/3$  reproduces the  $N \leq 4$  list in Appendix A. At  $c = 1/2$ , collided labels make his denominator vanish to higher order (for instance  $\Delta_{\langle 2,1 \rangle} = \Delta_{\langle 1,3 \rangle} = \frac{1}{2}$  at order three), and the specialized numerators  $K_N(\Delta, d, -4/3)$  acquire the compensating zeros, so that the actual pole multiplicities are the  $m_{n,N}$  of Definition 7.3. The present paper upgrades these closed low-order expressions to an all-order fixed- $c$  statement with a termination and equality proof.

## 11. Irreducible Ising specialization

**Proposition 11.1** (*Identity insertion in the Verma recursion*). For  $d = 0$ ,

$$H_{\Delta}^{(1/2)}(0 \mid q) = 1$$

as a rational function of  $\Delta$ . Thus specialization of the meromorphic Verma block at a degenerate weight is not the irreducible Ising trace; the latter requires taking the quotient by the radical.

*Proof.* In Definition 6.1, the factor with  $k = l = 1$  in  $\mathcal{P}_{r,s}(b, 0)$  is

$$\frac{(b + b^{-1})^2 - (b + b^{-1})^2}{4} = 0.$$

Hence every generic residue  $R_a(t)$  vanishes identically at  $d = 0$ . Definitions 7.1 and 7.3 then give  $\tilde{A}_{n,k,N}(0) = 0$ , hence  $A_{n,k,N}(0) = 0$ , for every  $N \geq 1$ . Equivalently, in the inverse-Shapovalov definition the identity insertion has  $\rho_N = G_N$ , so  $F_N = p(N)$  and eta reduction gives  $H = 1$ . ■

**Definition 11.2** (Eta-reduced irreducible identity block). For  $h \in \{0, 1/2, 1/16\}$ , let  $L(1/2, h)$  be the irreducible quotient of  $M(1/2, h)$ , and define

$$\hat{H}_h(q) = \prod_{m \geq 1} (1 - q^m) \sum_{N \geq 0} \dim L(1/2, h)_{h+N} q^N.$$

**Theorem 11.3** (*The three Ising characters*). The quotient identity blocks of Definition 11.2 are

$$\hat{H}_0(q) = \sum_{j \in \mathbb{Z}} \left( q^{((24j+1)^2-1)/48} - q^{((24j+7)^2-1)/48} \right),$$

$$\hat{H}_{1/2}(q) = \sum_{j \in \mathbb{Z}} \left( q^{((24j+5)^2-25)/48} - q^{((24j+11)^2-25)/48} \right),$$

and

$$\widehat{H}_{1/16}(q) = \sum_{j \in \mathbb{Z}} \left( q^{((24j-2)^2-4)/48} - q^{((24j+10)^2-4)/48} \right).$$

Consequently,

$$\chi_h(q) = q^{h-1/48} \frac{\widehat{H}_h(q)}{\prod_{m \geq 1} (1 - q^m)}$$

are the vacuum, energy, and spin Ising characters. Their eta-reduced beginnings are

$$\begin{aligned} \widehat{H}_0 &= 1 - q - q^6 + q^{11} + q^{13} - q^{20} + \dots, \\ \widehat{H}_{1/2} &= 1 - q^2 - q^3 + q^7 + q^{17} - q^{25} - q^{28} + \dots, \\ \widehat{H}_{1/16} &= 1 - q^2 - q^4 + q^{10} + q^{14} - q^{24} - q^{30} + \dots. \end{aligned}$$

*Proof.* The Ising model is the minimal series  $c_{3,4} = 1/2$ . The Class  $R^+$ , Case 1+, BGG resolution in [HK11, Theorem 6.9] is an exact complex of Verma modules. Taking its Euler–Poincaré character gives, for an interior label  $(r, s)$ ,

$$\chi_{r,s}(q) = \frac{1}{\eta(q)} \sum_{j \in \mathbb{Z}} \left( q^{(24j+4r-3s)^2/48} - q^{(24j+4r+3s)^2/48} \right).$$

Use  $(r, s) = (1, 1), (2, 1), (1, 2)$ , respectively. Since

$$h_{r,s} = \frac{(4r-3s)^2-1}{48}, \quad \eta(q) = q^{1/24} \prod_{m \geq 1} (1 - q^m),$$

factoring out  $q^{h_{r,s}-1/48}$  gives exactly the three displayed  $\widehat{H}$ -series. For any fixed power of  $q$ , only finitely many integers  $j$  contribute, so the Euler–Poincaré sums are coefficientwise finite. Direct evaluation of the exponents through degree 30 gives the final three lines. ■

*Remark 11.4* (Character replay). The three BGG numerators and the resulting quotient dimensions through level 30 are replayed by `tools/audit_ising_characters.py` using exact integer arithmetic, and are checked there against an independent free-fermion count: the spin module is counted by partitions into distinct positive integers, and the vacuum and energy modules by partitions of  $2N$  (respectively  $2N+1$ ) into distinct odd parts with an even (respectively odd) number of parts.

## 12. Scope and outlook

Theorem 4.4 is a coefficientwise theorem for the meromorphic Verma-module torus block at every fixed nonzero algebraic  $b_*$ . In particular, it applies to every minimal-model central charge; Proposition 3.2 supplies the corresponding collision classes. The Ising specialization is developed in greater detail because its classes, coefficient field, embedding diamonds, and exact local computations are especially transparent. The output is fixed at  $c_*$ , while the construction uses the exact formal deformation  $b = b_* + t$  to combine resonant generic-recursion terms. The method should therefore be distinguished both from numerical regularization and from a recursion formulated solely in intrinsic data of the specialized category.

At any prescribed level the label sets and recursion trees are finite. Lemmas 7.2 and 7.4 give an a priori Laurent jet budget, but do not claim a favorable complexity bound: exact rational simplification can grow rapidly with the level. The local Smith analysis explains the first simple and double poles, but an all-depth identification of every Smith layer with the Ising embedding diagram is not required for the coefficientwise confluence theorem. The level-ten principal part has been verified directly against the inverse-Shapovalov definition (Remark 9.6); an independent symbolic construction of the level-ten  $42 \times 42$  second crossing form itself would nevertheless be a useful additional audit.

Open directions include uniform complexity bounds and a representation- theoretic description of the resulting pole layers as  $p, p'$  vary, an intrinsic fixed-central-charge recursion that eliminates the formal deformation, and a corresponding recursion for irreducible minimal-model traces at degenerate internal weights. The BGG calculation of Section 11 is only a quotient consistency check and is logically separate from the confluence theorem.

## Code availability

All exact-arithmetic audit scripts cited in the text are distributed with this paper as ancillary files: the eight `tools/audit_*.py` scripts described in Remarks 6.2 and 9.6 and in the character-replay remark, together with the shared library modules `tools/exact_shapovalov.py` and `tools/direct_reconstruction.py` on which they depend. All computations use exact rational arithmetic (Python 3, tested with CPython 3.12, and SymPy 1.12; the pinned dependency is listed in `requirements-audit.txt`). Running `make audit` replays every check except the direct level-ten reconstruction, which the slower target `make audit-deep` adds. Exact commands, expected success markers, and SHA-256 hashes of the audit sources are recorded in `AUDIT_MANIFEST.md`; the complete package is part of the submitted ancillary material and will additionally be deposited in a citable archive.

## A. Expanded polynomial data

The factored polynomial entering the level-seven principal part is

$$\begin{aligned} \Theta_7(d) = & - \frac{d^2(d-1)^2(2d-15)(2d-7)(2d-5)(2d-1)(3d-14)(3d-5)}{2164322160000} \\ & \times (84001d^4 - 3376370d^3 + 44478887d^2 \\ & - 209562838d + 205096320). \end{aligned} \quad (\text{A.1})$$

The corresponding level-ten polynomial is

$$\begin{aligned} \Theta_{10}(d) = & - \frac{d^2(d-1)^2(2d-7)^2(2d-1)^2(3d-14)(3d-5)(6d-55)(6d-1)}{552735279561465000000} \\ & \times (424640464d^8 - 41506307104d^7 + 1664502727528d^6 \\ & - 35552058801520d^5 + 438148123547677d^4 - 3142854705332170d^3 \\ & + 12488401942129875d^2 - 23870935242348750d + 14566042731240000). \end{aligned} \quad (\text{A.2})$$

The complete nonzero partial-fraction output through level four is

$$\begin{aligned}
A_{1,1,1} &= \frac{1}{2} d(d-1), \\
A_{1,1,2} &= \frac{1}{4} d^2(d-1)^2, \\
A_{2,1,2} &= -\frac{1}{7} d(d-1)(2d-7)(2d-1), \\
A_{5,1,2} &= \frac{1}{28} d(d-1)(3d-14)(3d-5), \\
A_{1,1,3} &= \frac{1}{120} d^2(d-1)^2(17d^2 - 113d + 236), \\
A_{2,1,3} &= -\frac{8}{231} d^2(d-1)^2(2d-7)(2d-1), \\
A_{5,1,3} &= -\frac{1}{84} d(d-1)(d^4 - 50d^3 + 311d^2 - 550d + 210), \\
A_{9,1,3} &= \frac{3}{3080} d(d-11)(d-6)(d-1)(3d-14)(3d-5), \\
A_{1,1,4} &= -\frac{1}{1440} d^2(d-1)^2(31d^4 - 830d^3 + 5567d^2 - 13408d + 8460), \\
A_{2,1,4} &= \frac{1}{40425} d(d-1)(2d-7)(2d-1) \\
&\quad \times (248d^4 - 5752d^3 + 25274d^2 - 33630d + 5775), \\
A_{5,1,4} &= -\frac{1}{61152} d^2(d-1)^2(149d^4 - 2506d^3 - 2679d^2 + 70508d - 106540), \\
A_{8,1,4} &= -\frac{2}{20475} d(d-13)(d-6)(d-1)(2d-15)(2d-7)(2d-5)(2d-1), \\
A_{9,1,4} &= \frac{9}{86240} d^2(d-11)(d-6)(d-1)^2(3d-14)(3d-5), \\
A_{13,1,4} &= \frac{1}{184800} d(d-20)(d-13)(d-11)(d-6)(d-1)(3d-14)(3d-5).
\end{aligned} \tag{A.3}$$

## B. The level-four Gram matrix at $c = 1/2$

In the ordered PBW basis  $(L_{-4}, L_{-3}L_{-1}, L_{-2}^2, L_{-2}L_{-1}^2, L_{-1}^4)|\Delta\rangle$  of Proposition 5.9, direct Virasoro commutation at  $c = 1/2$  gives the symmetric matrix  $G_4 = (C \ C')$  with column blocks

$$\begin{aligned}
C &= \begin{pmatrix} 8\Delta + \frac{5}{2} & 14\Delta & 24\Delta + \frac{3}{2} \\ 14\Delta & 12\Delta^2 + 14\Delta & 30\Delta \\ 24\Delta + \frac{3}{2} & 30\Delta & 32\Delta^2 + 36\Delta + \frac{17}{8} \\ 36\Delta & 40\Delta^2 + 20\Delta & 48\Delta^2 + 51\Delta \\ 120\Delta & 192\Delta^2 + 48\Delta & 216\Delta^2 + 144\Delta \end{pmatrix}, \\
C' &= \begin{pmatrix} 36\Delta & 120\Delta \\ 40\Delta^2 + 20\Delta & 192\Delta^2 + 48\Delta \\ 48\Delta^2 + 51\Delta & 216\Delta^2 + 144\Delta \\ 32\Delta^3 + 118\Delta^2 + 33\Delta & 288\Delta^3 + 336\Delta^2 + 96\Delta \\ 288\Delta^3 + 336\Delta^2 + 96\Delta & 384\Delta^4 + 1152\Delta^3 + 1056\Delta^2 + 288\Delta \end{pmatrix}.
\end{aligned}$$

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